

BFS Traversal

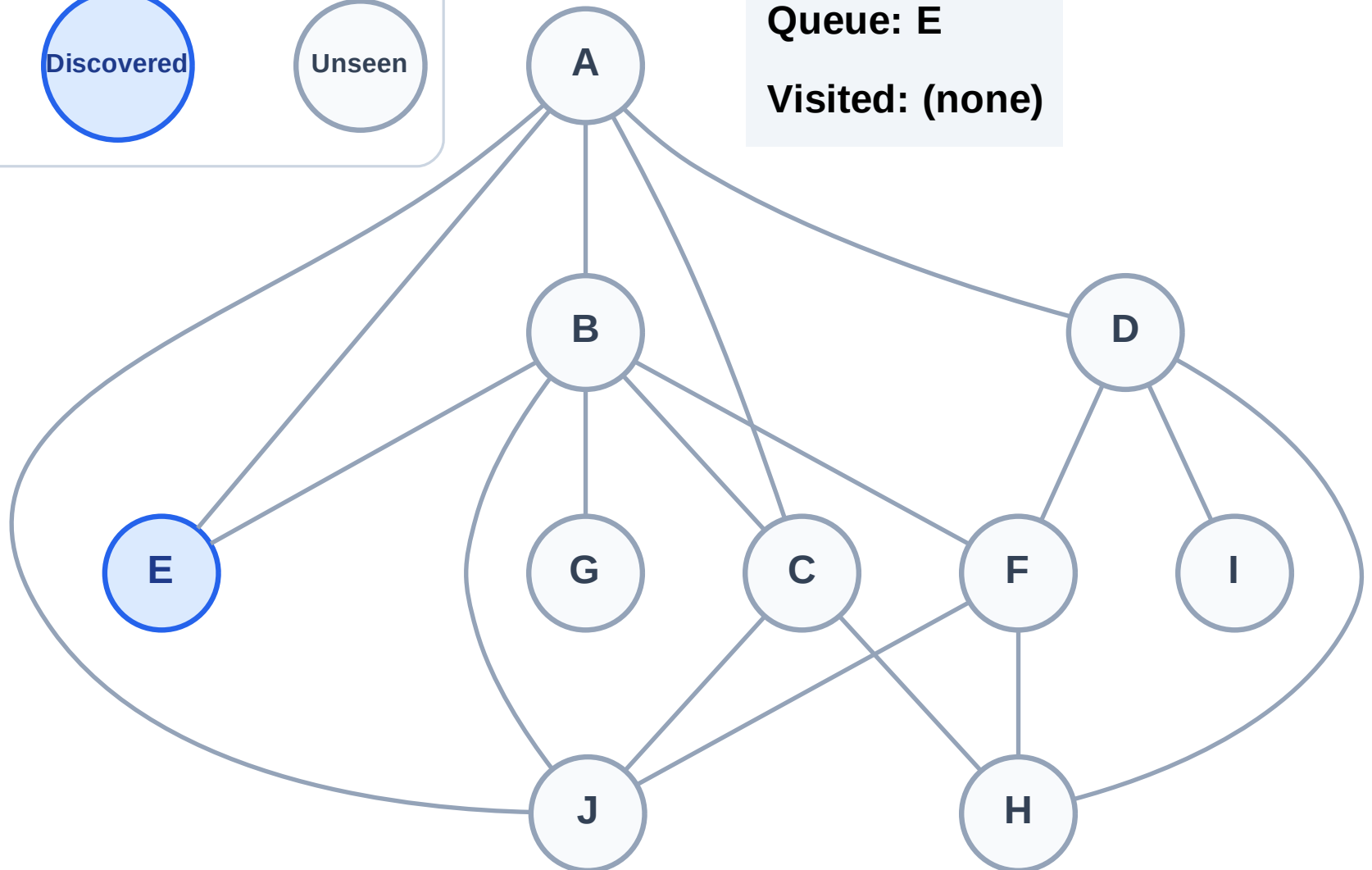
Step 1: Start at E

Legend



Queue: E

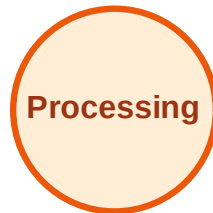
Visited: (none)



BFS Traversal

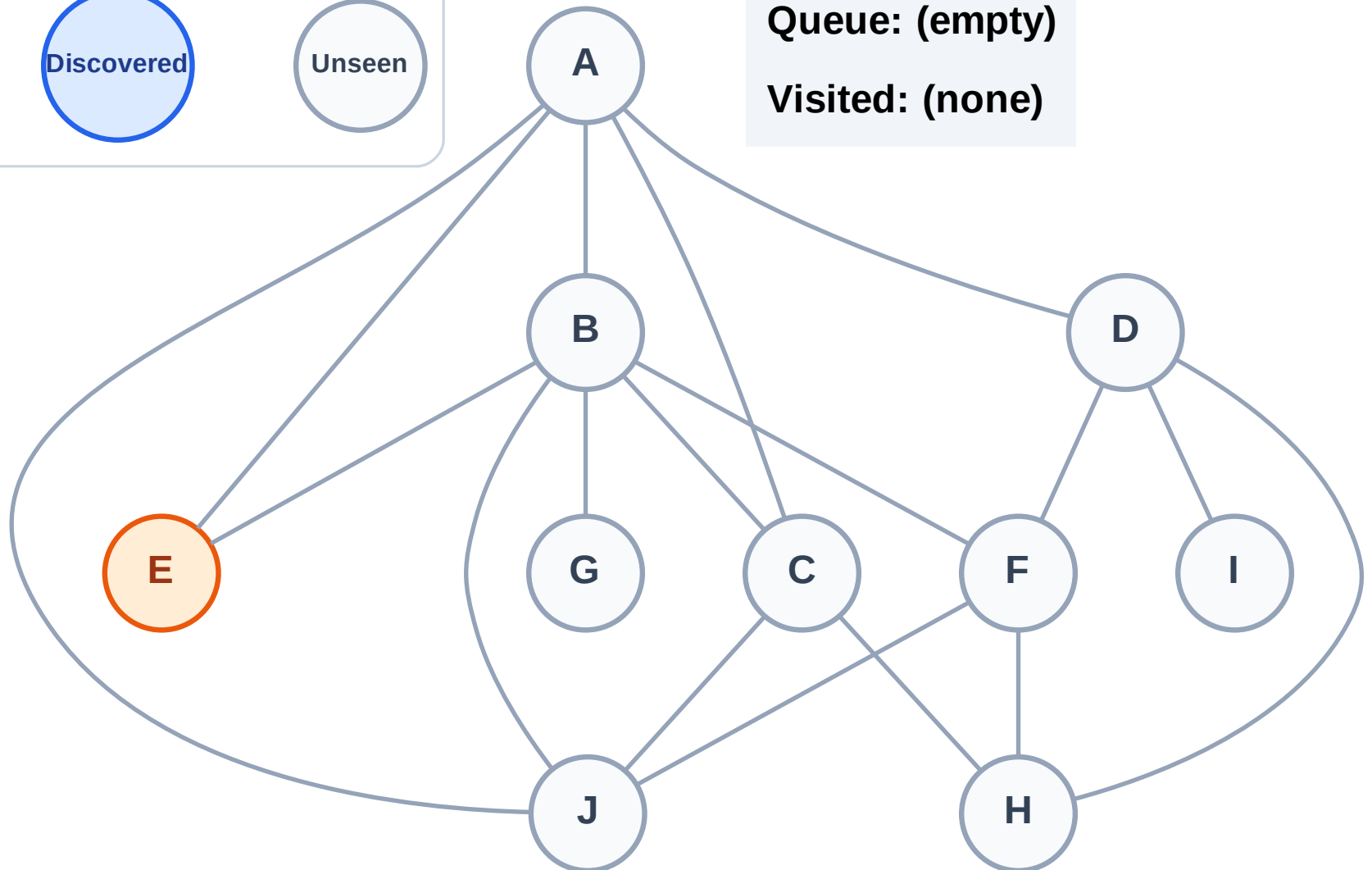
Step 2: Process node E

Legend



Queue: (empty)

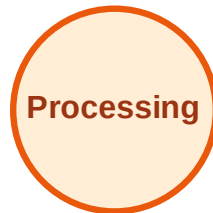
Visited: (none)



BFS Traversal

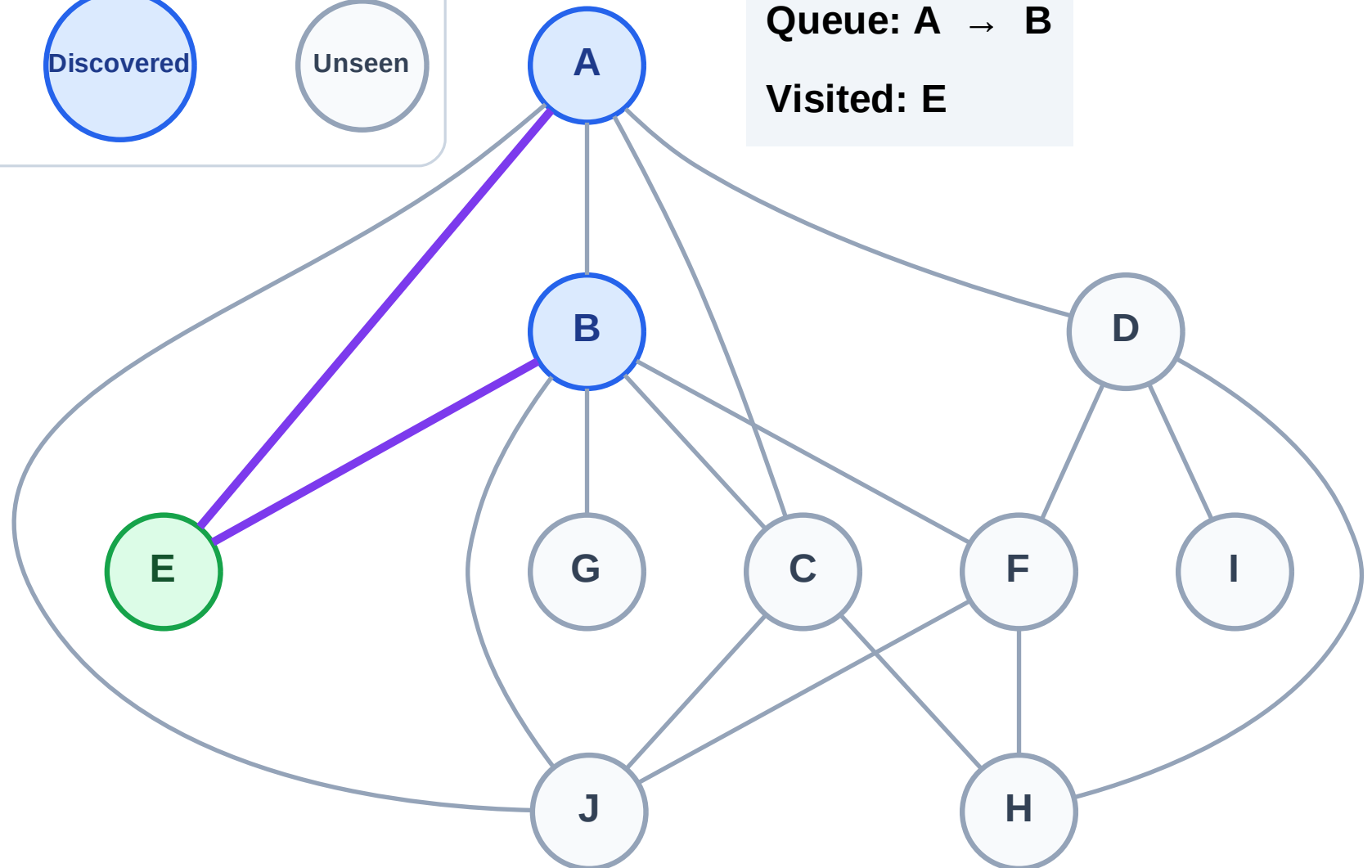
Step 3: Discover A, B from E

Legend



Queue: A → B

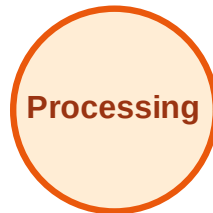
Visited: E



BFS Traversal

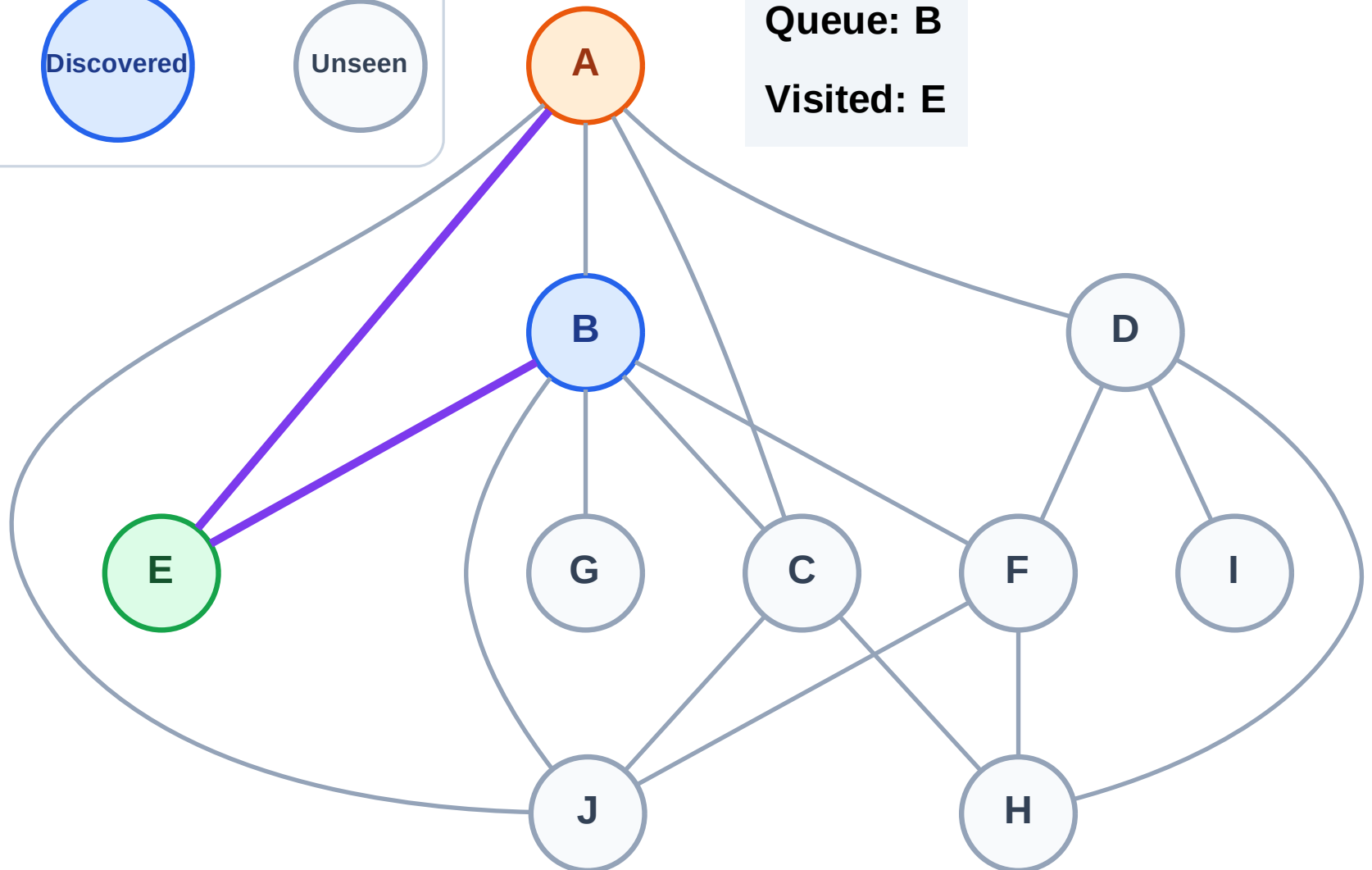
Step 4: Process node A

Legend



Queue: B

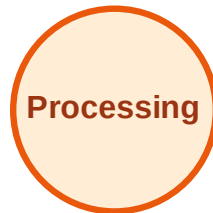
Visited: E



BFS Traversal

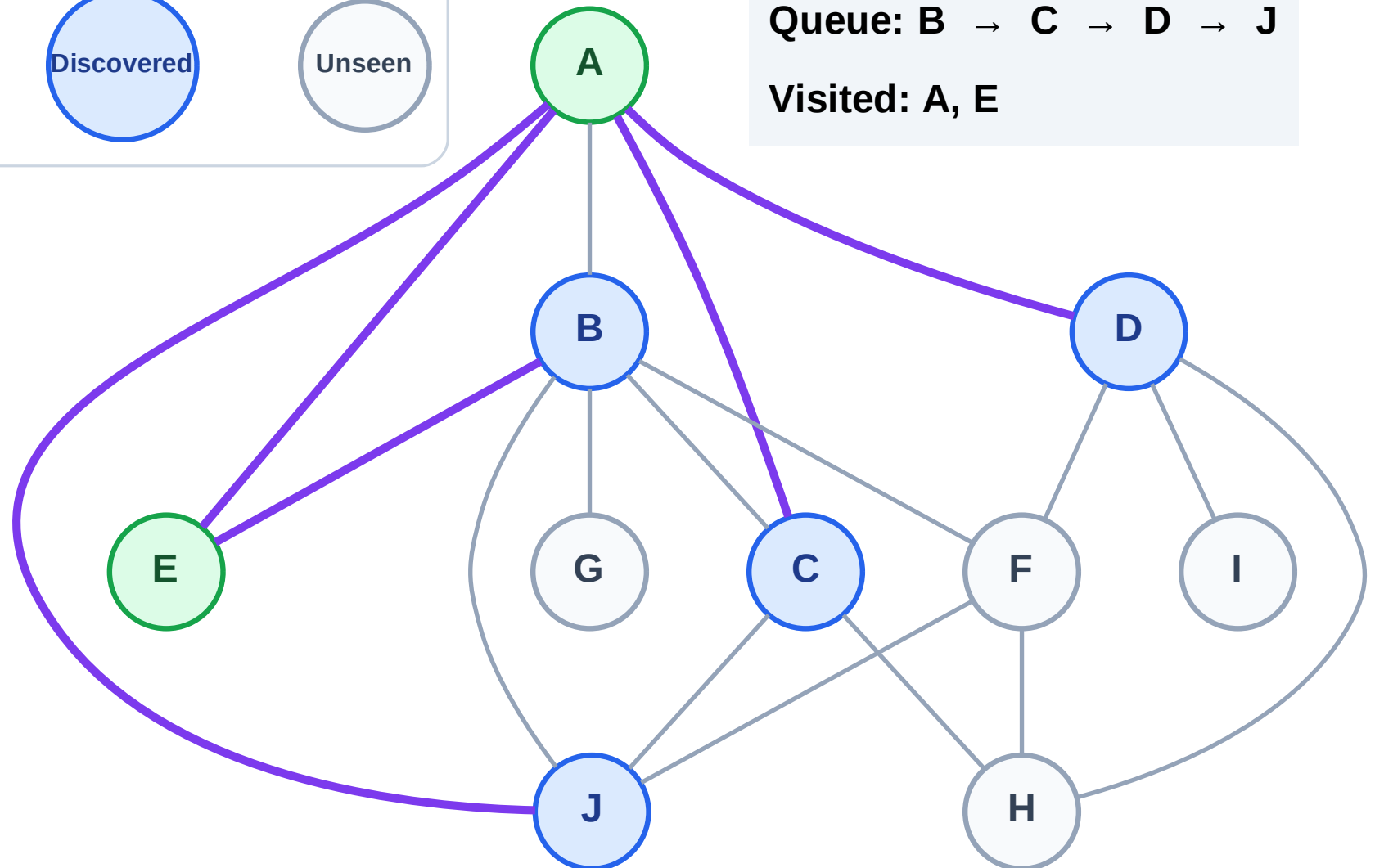
Step 5: Discover C, D, J from A

Legend



Queue: B → C → D → J

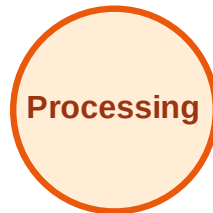
Visited: A, E



BFS Traversal

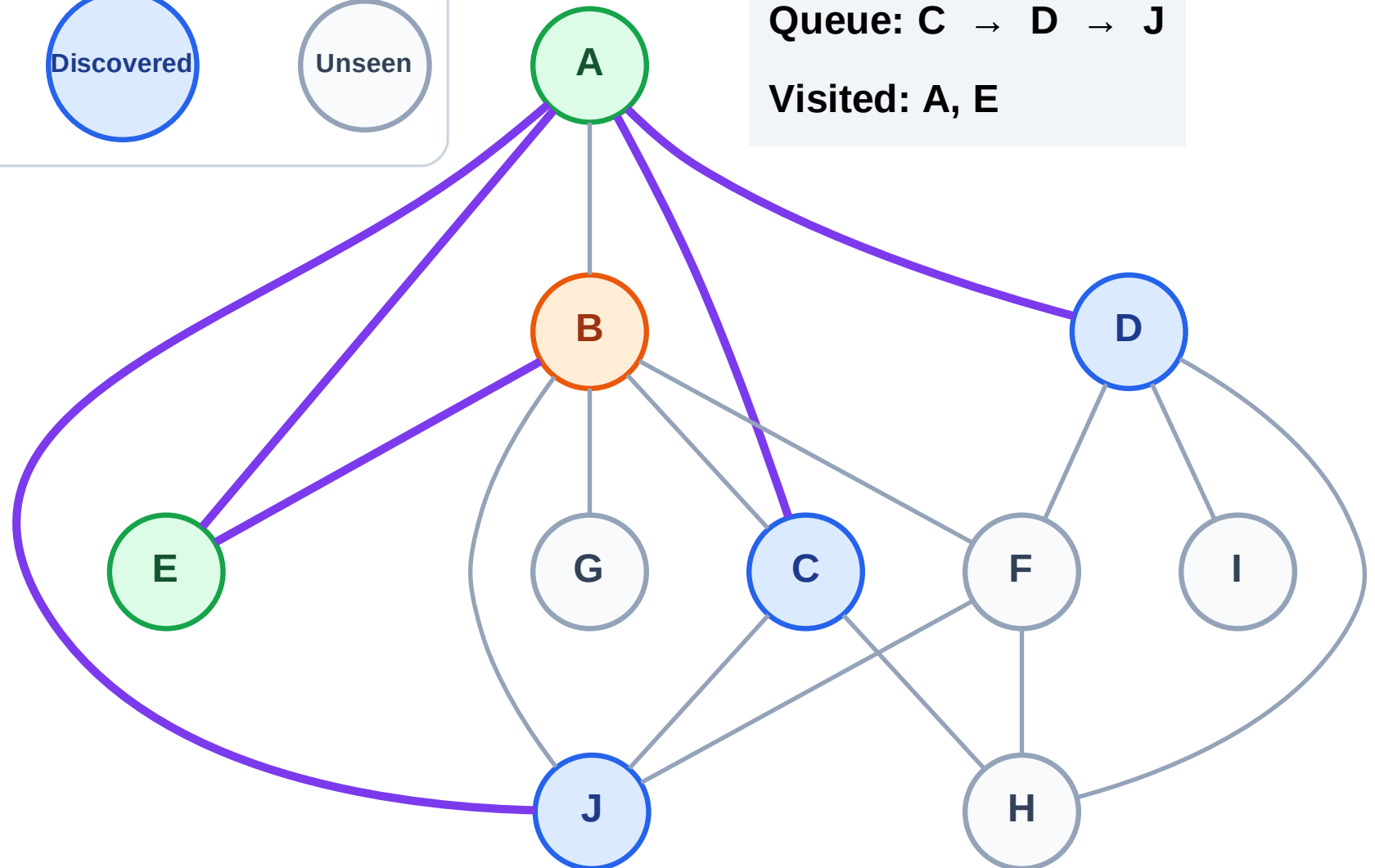
Step 6: Process node B

Legend



Queue: C → D → J

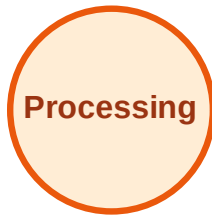
Visited: A, E



BFS Traversal

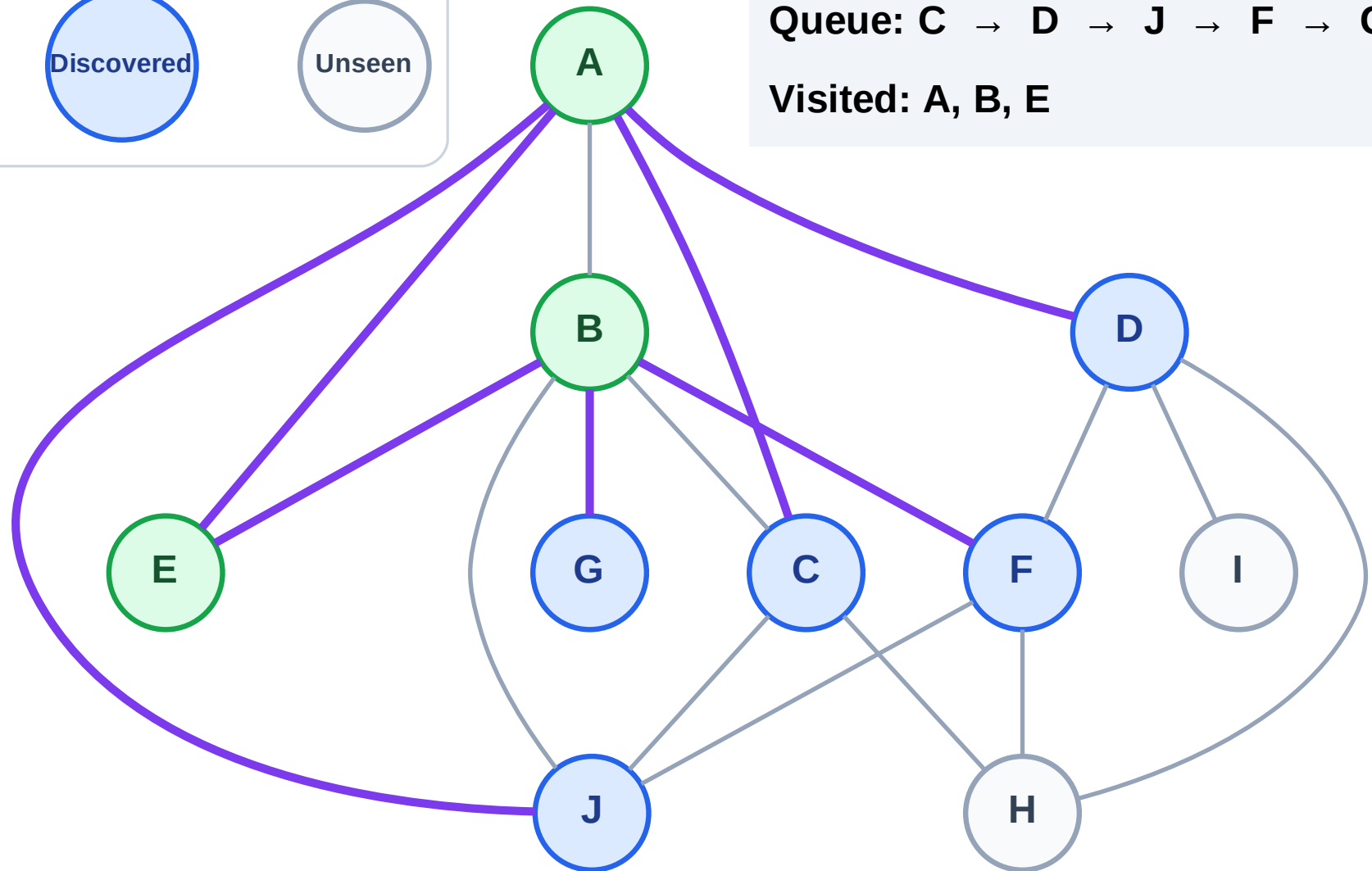
Step 7: Discover F, G from B

Legend



Queue: C → D → J → F → G

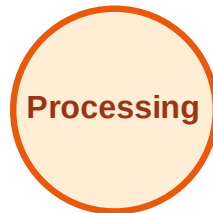
Visited: A, B, E



BFS Traversal

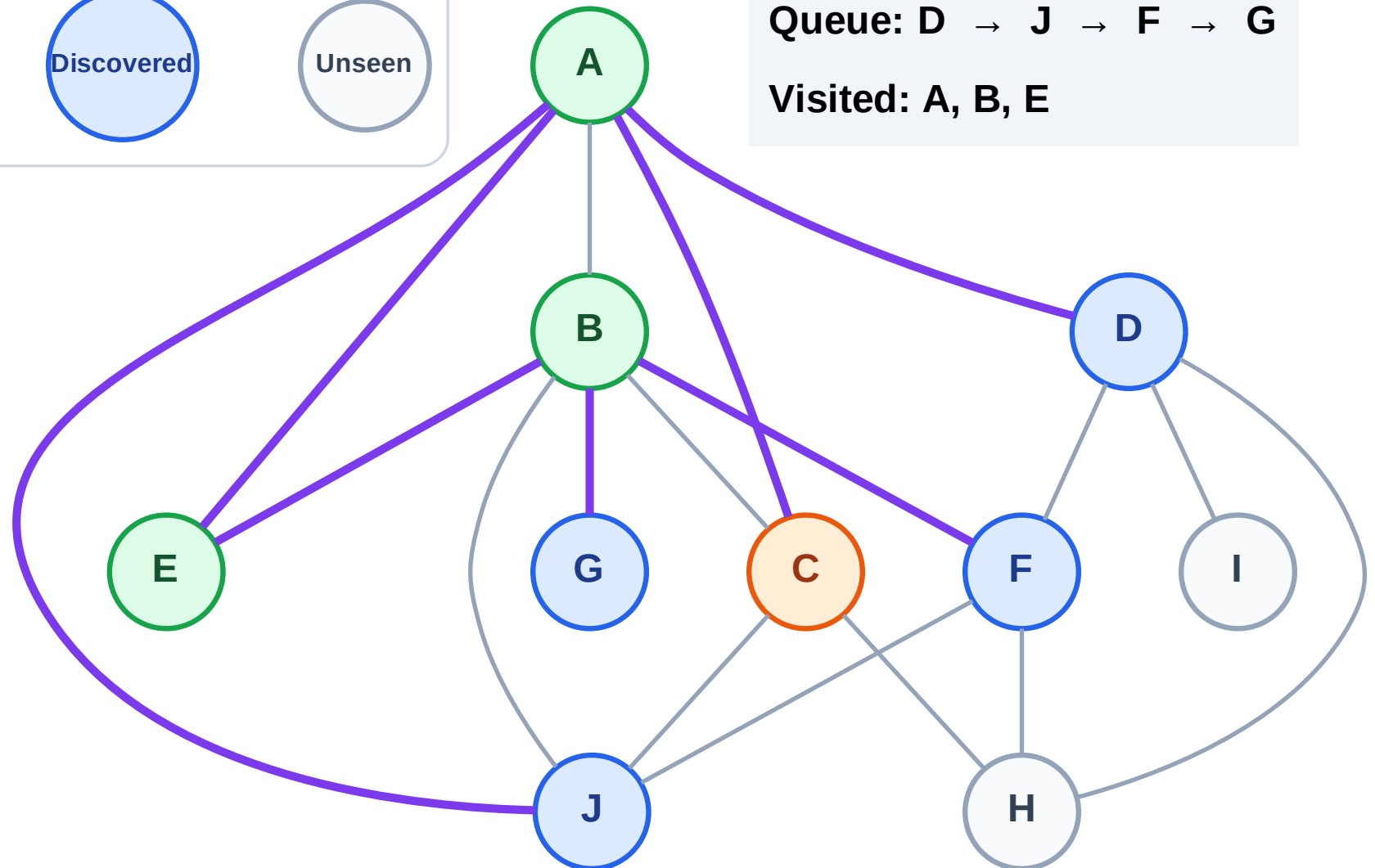
Step 8: Process node C

Legend



Queue: D → J → F → G

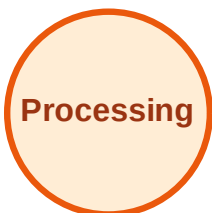
Visited: A, B, E



BFS Traversal

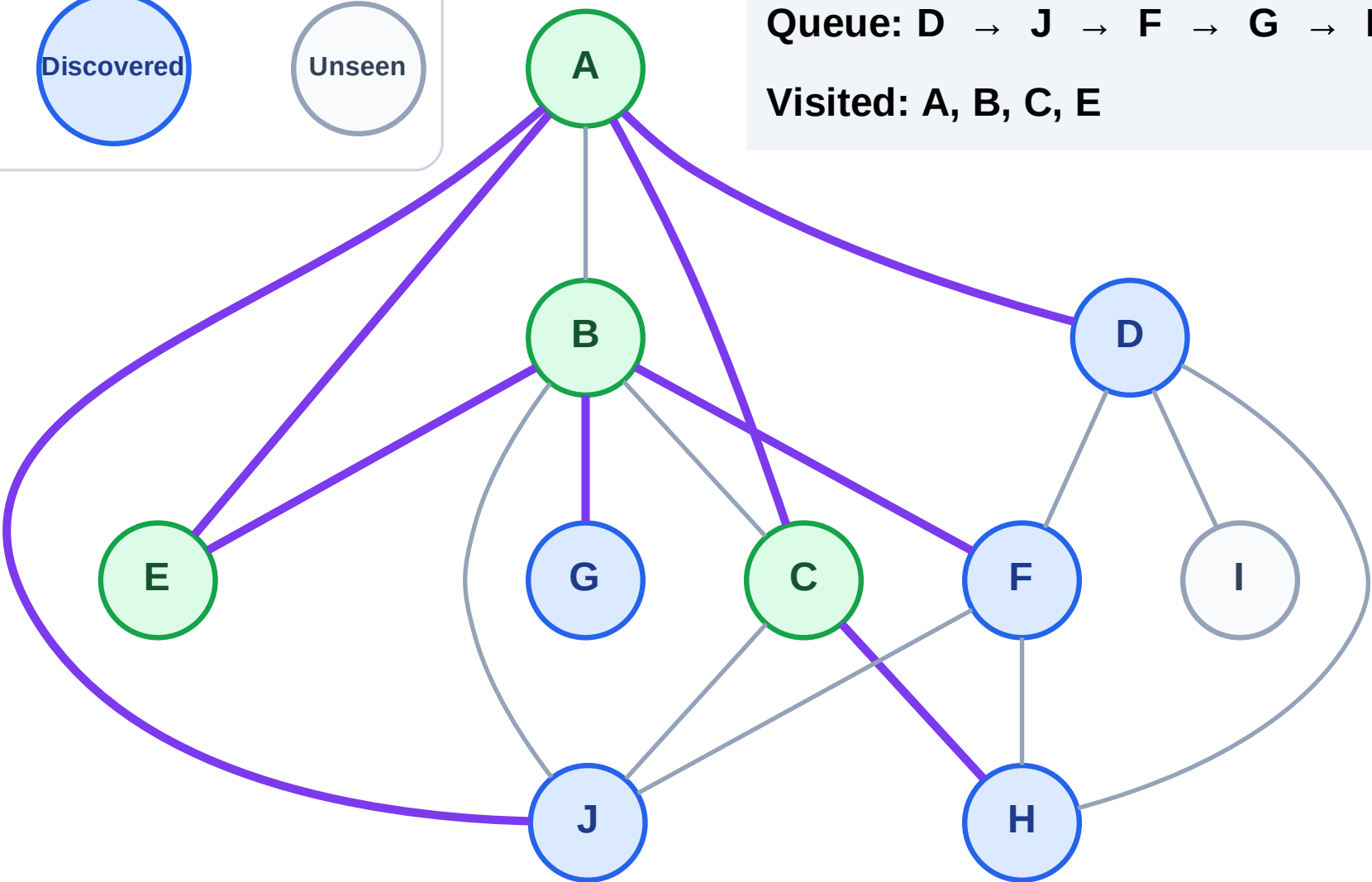
Step 9: Discover H from C

Legend



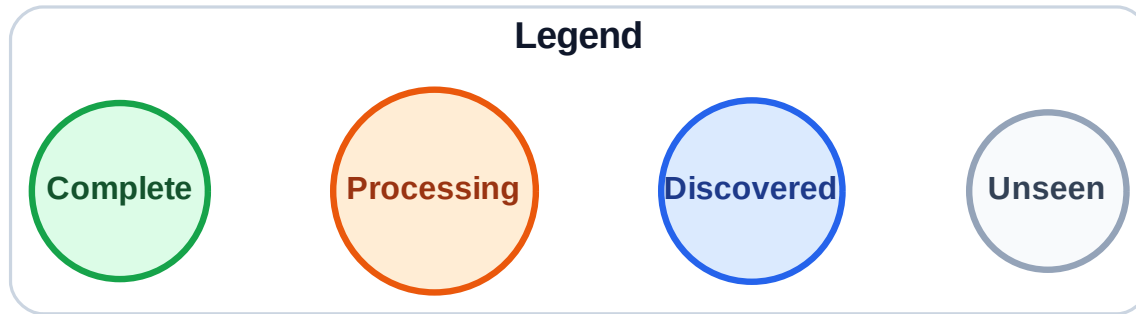
Queue: D → J → F → G → H

Visited: A, B, C, E



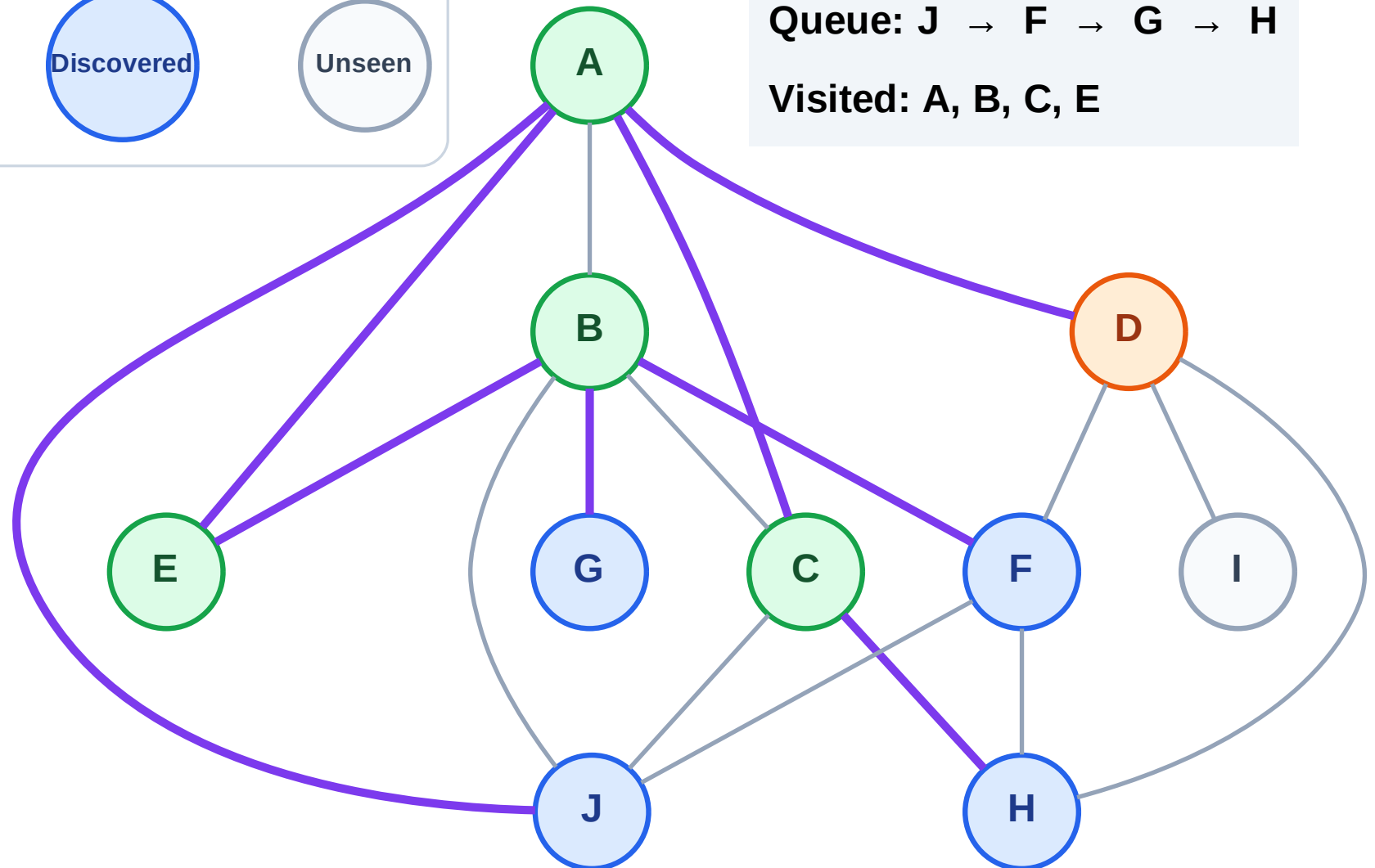
Step 10: Process node D

Step 10: Process node D



Queue: J → F → G → H

Visited: A, B, C, E



BFS Traversal

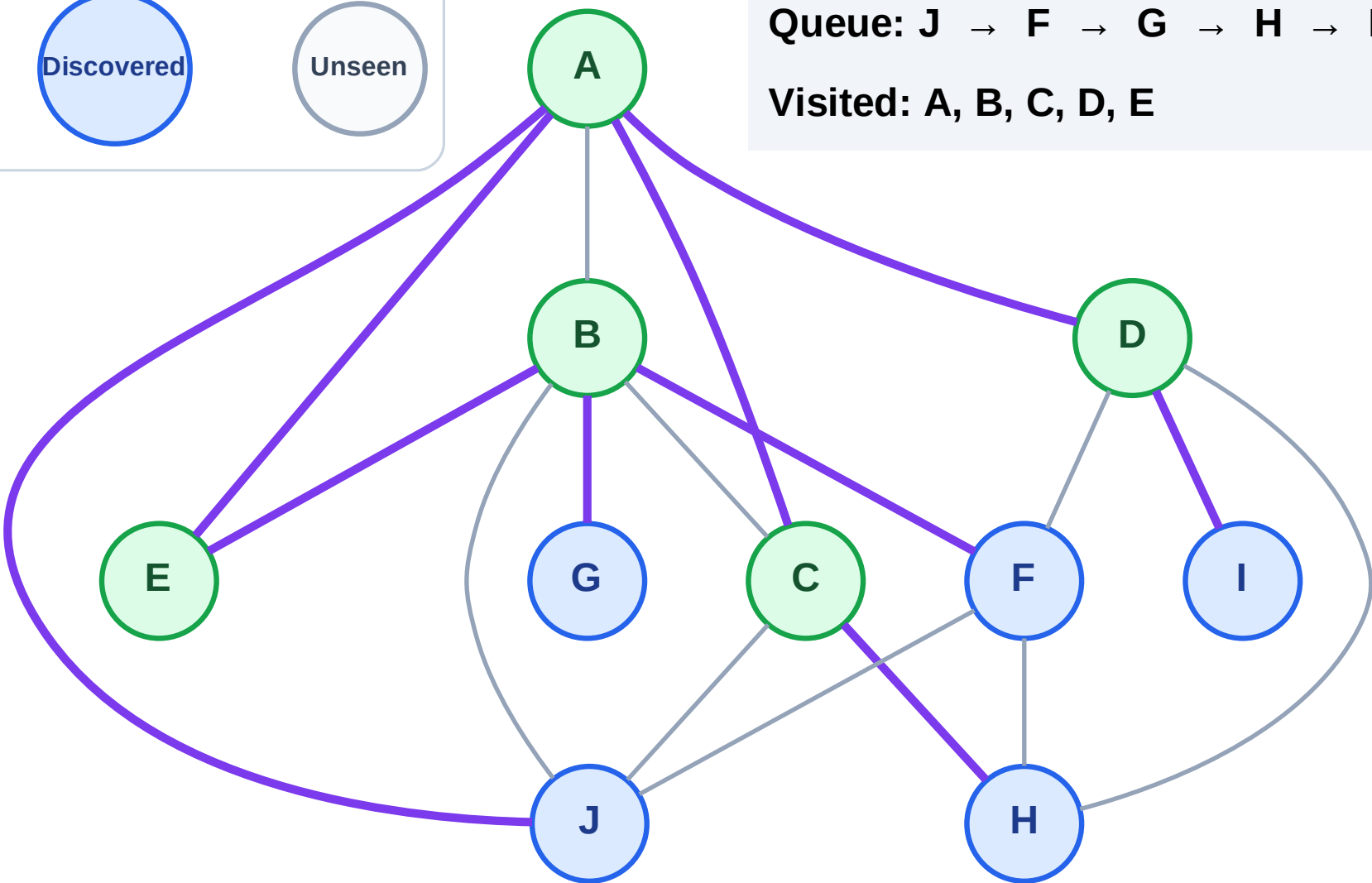
Step 11: Discover I from D

Legend



Queue: J → F → G → H → I

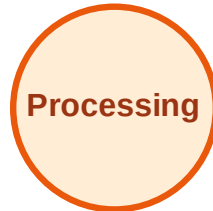
Visited: A, B, C, D, E



BFS Traversal

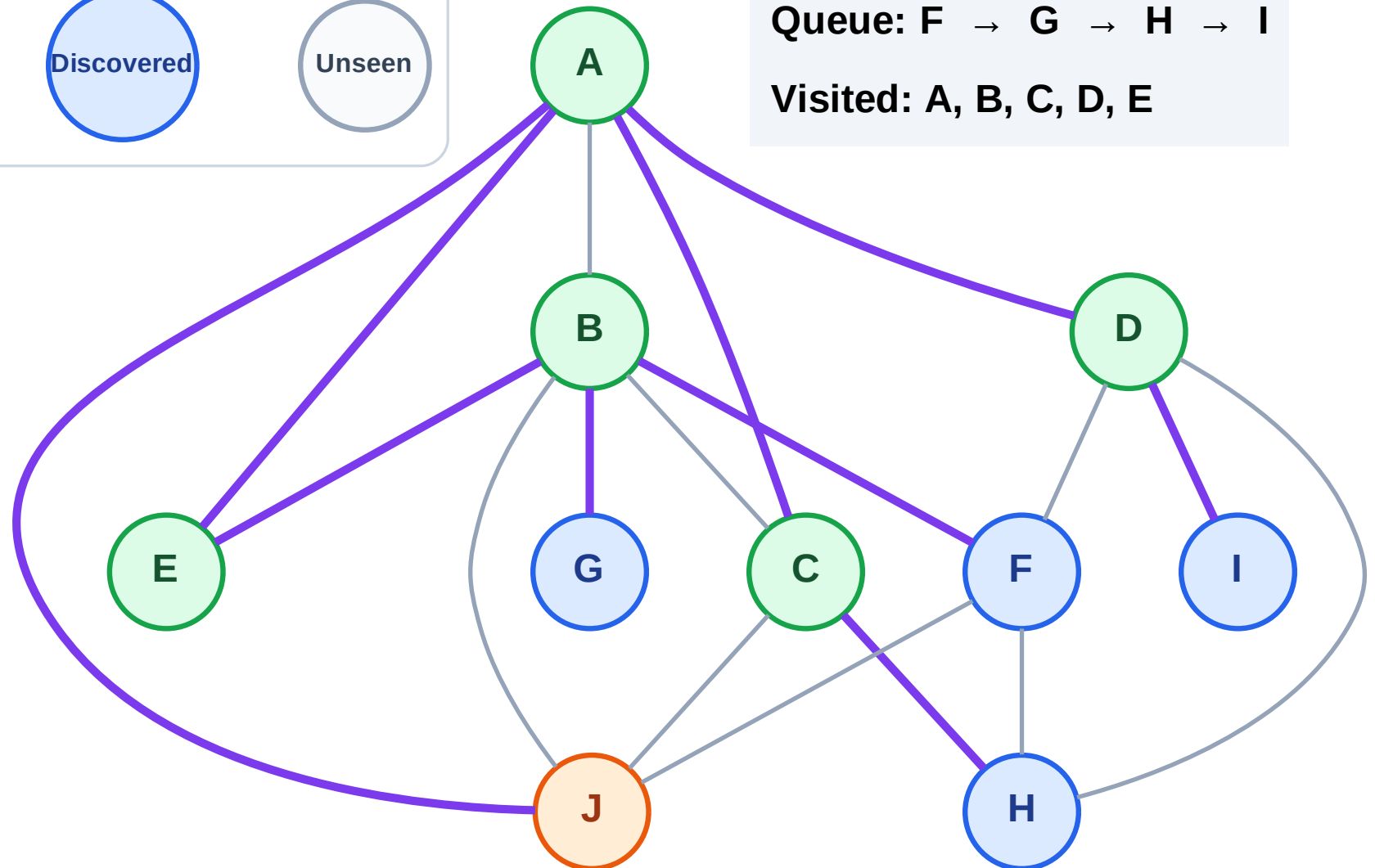
Step 12: Process node J

Legend



Queue: F → G → H → I

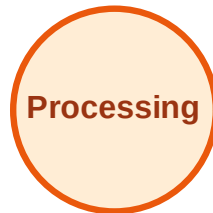
Visited: A, B, C, D, E



BFS Traversal

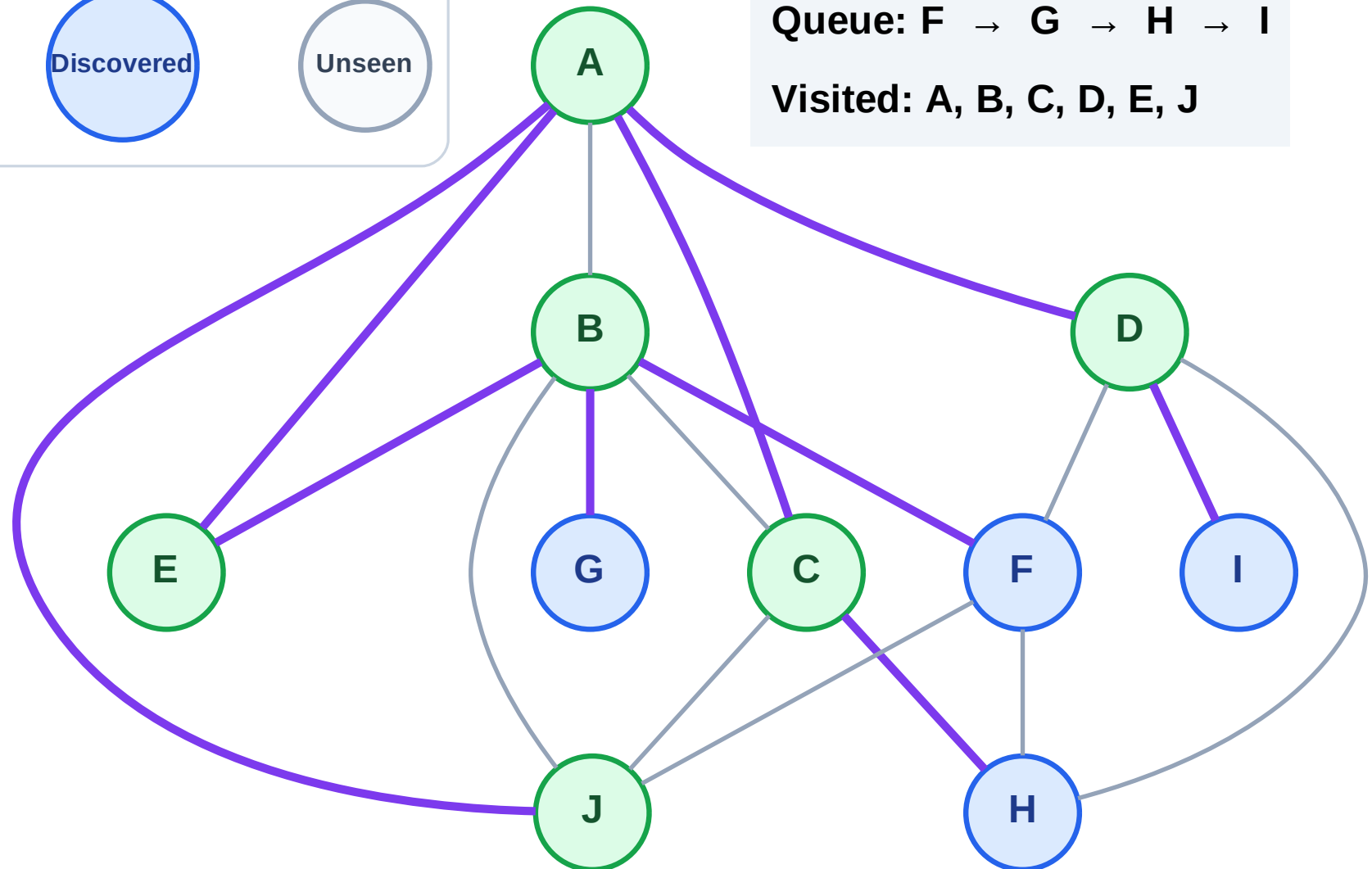
Step 13: No new neighbors from J

Legend



Queue: F → G → H → I

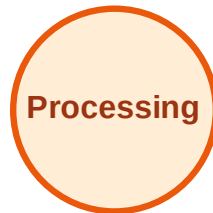
Visited: A, B, C, D, E, J



BFS Traversal

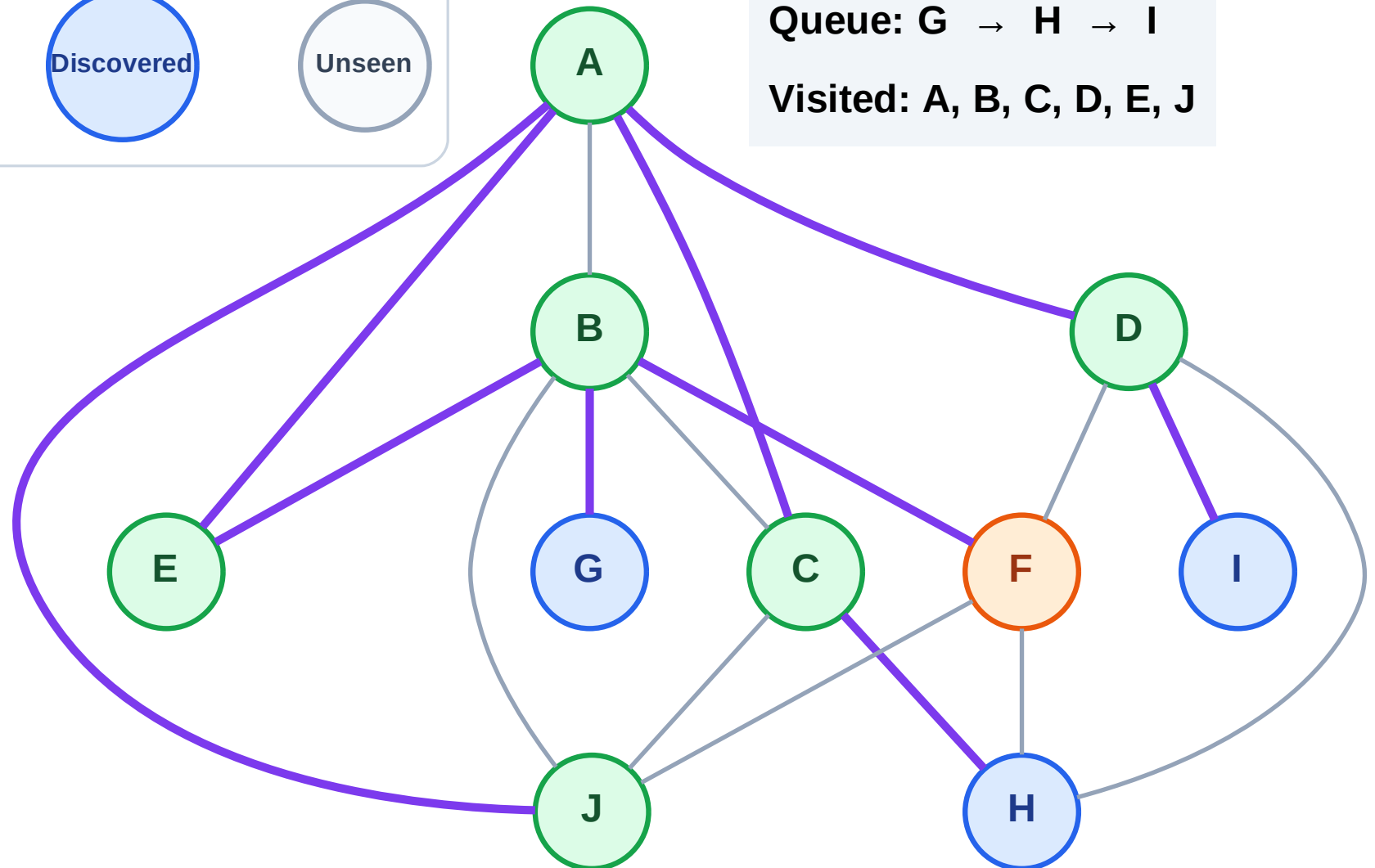
Step 14: Process node F

Legend



Queue: G → H → I

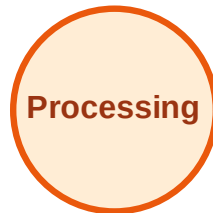
Visited: A, B, C, D, E, J



BFS Traversal

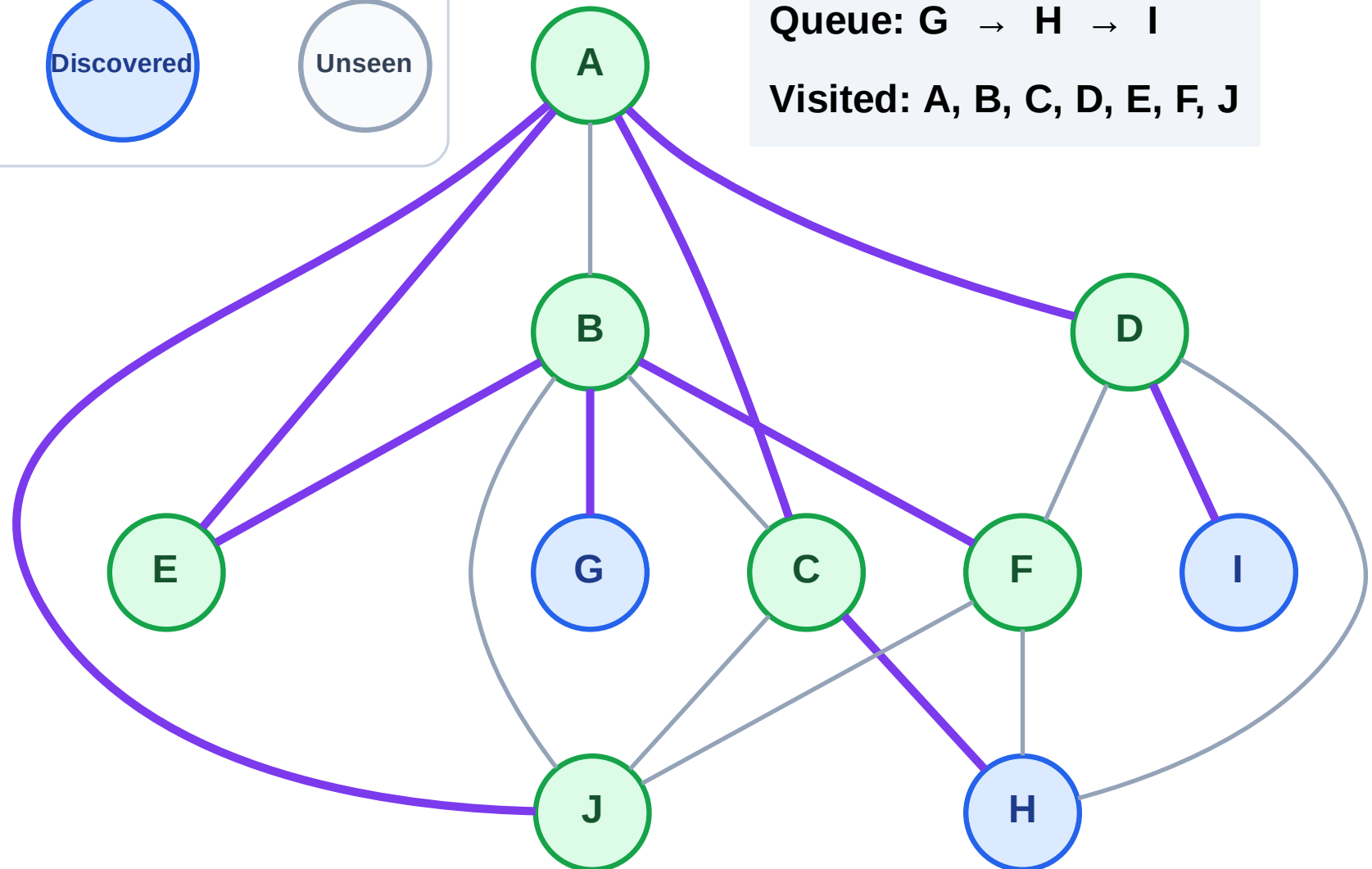
Step 15: No new neighbors from F

Legend



Queue: G → H → I

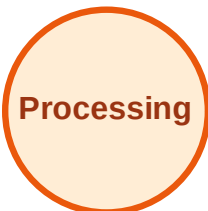
Visited: A, B, C, D, E, F, J



BFS Traversal

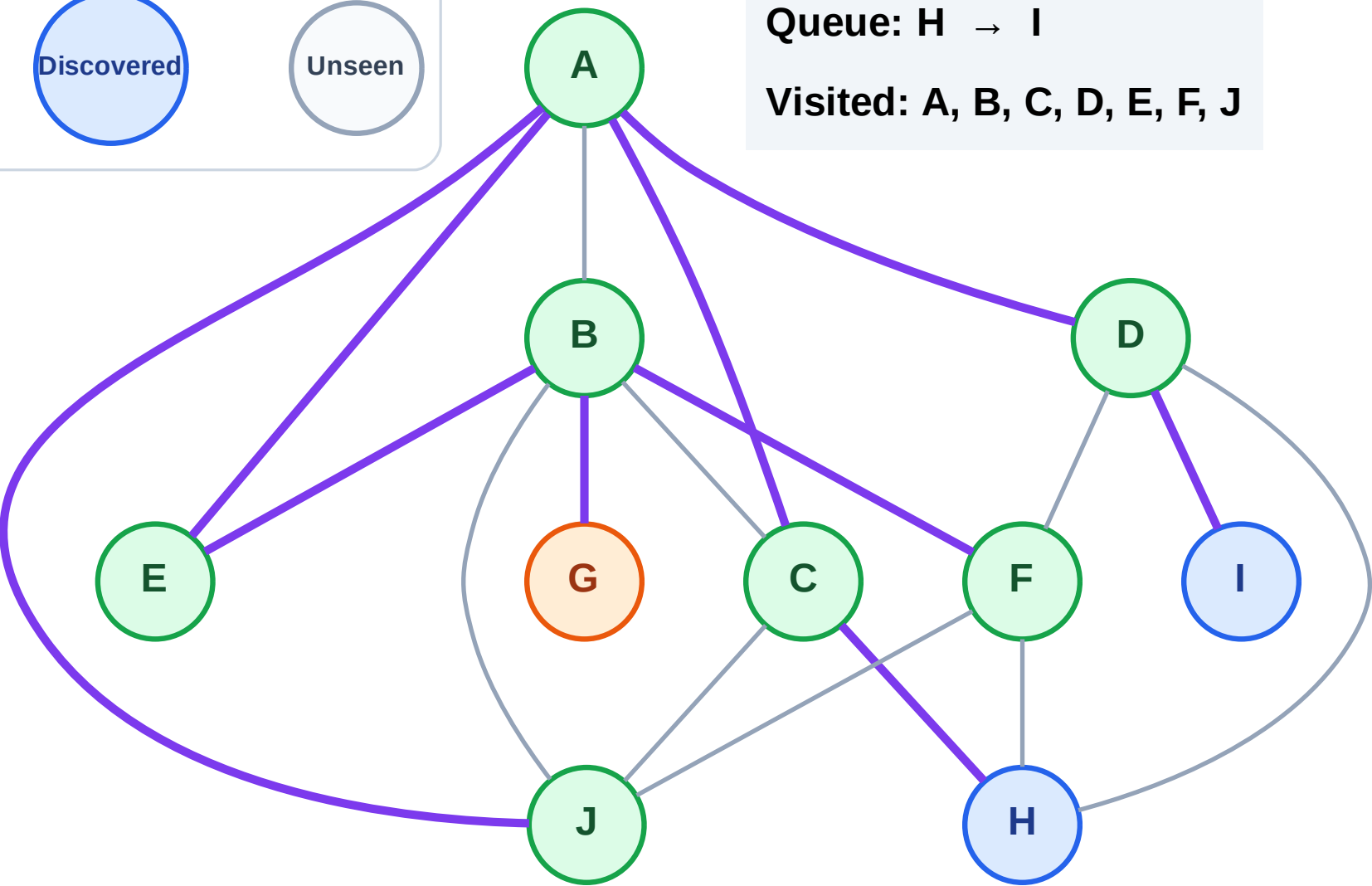
Step 16: Process node G

Legend



Queue: H → I

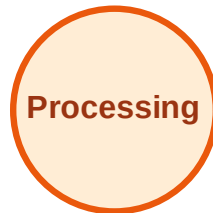
Visited: A, B, C, D, E, F, J



BFS Traversal

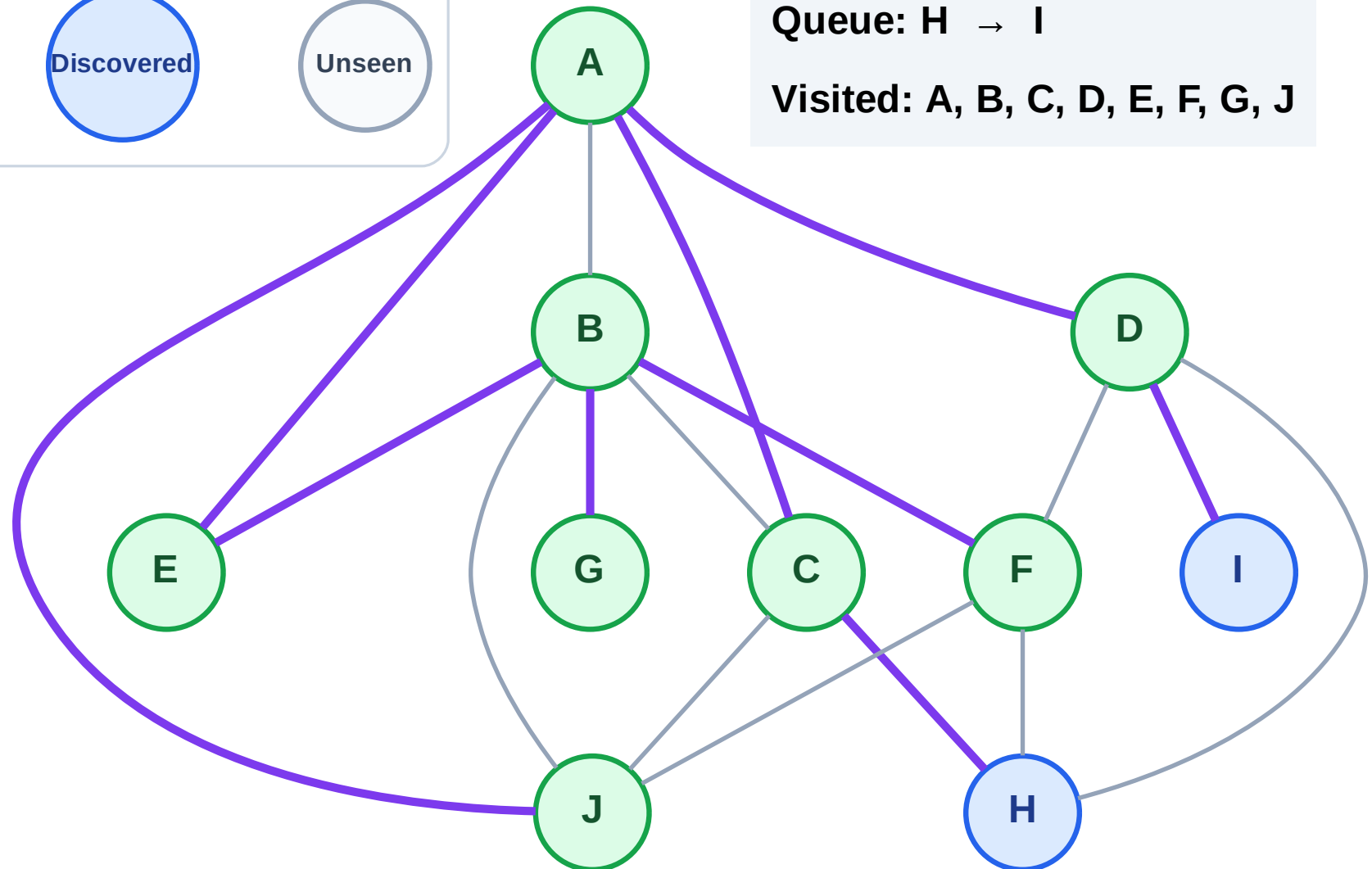
Step 17: No new neighbors from G

Legend



Queue: H → I

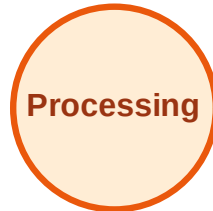
Visited: A, B, C, D, E, F, G, J



BFS Traversal

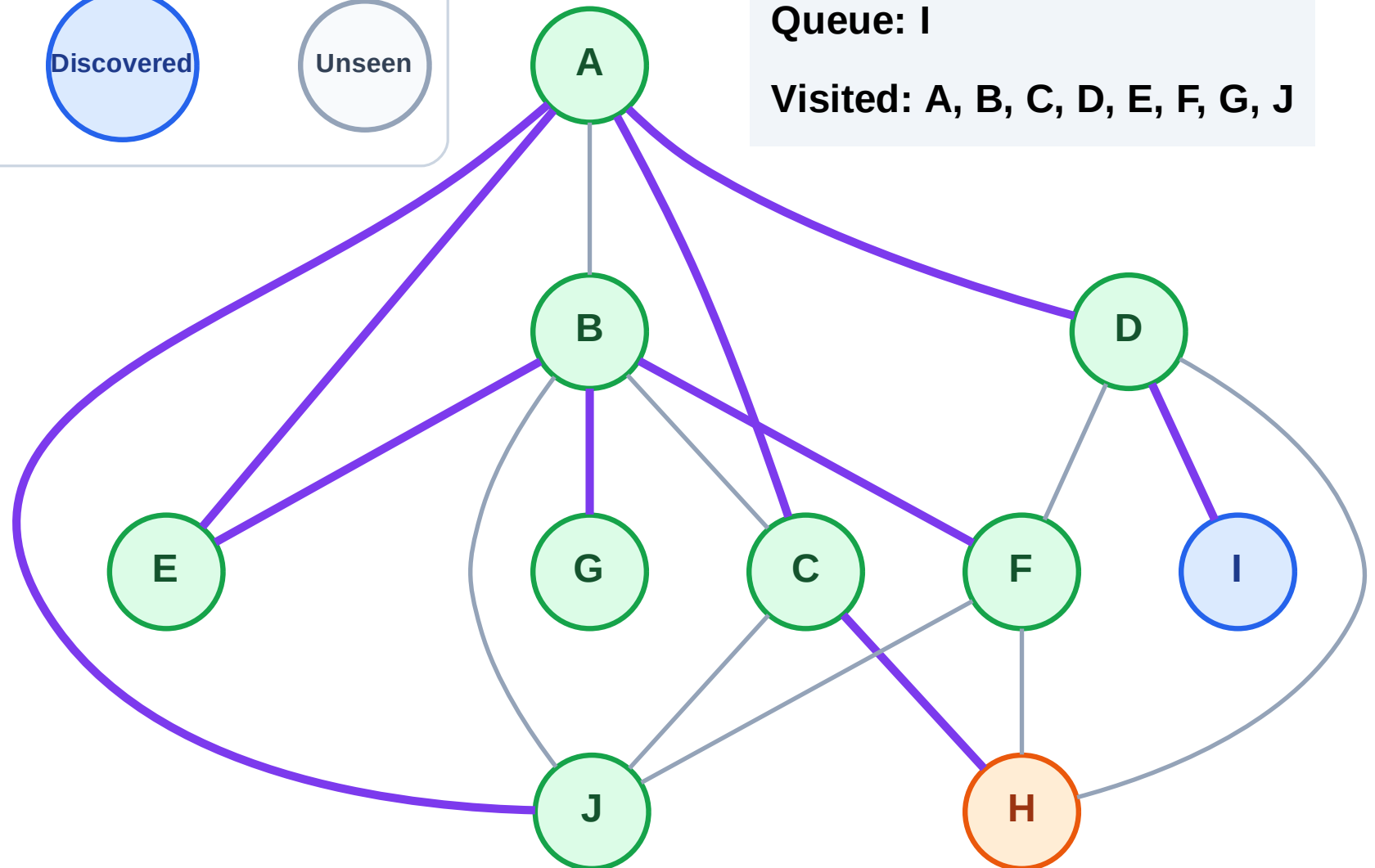
Step 18: Process node H

Legend



Queue: I

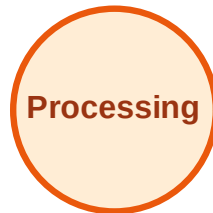
Visited: A, B, C, D, E, F, G, J



BFS Traversal

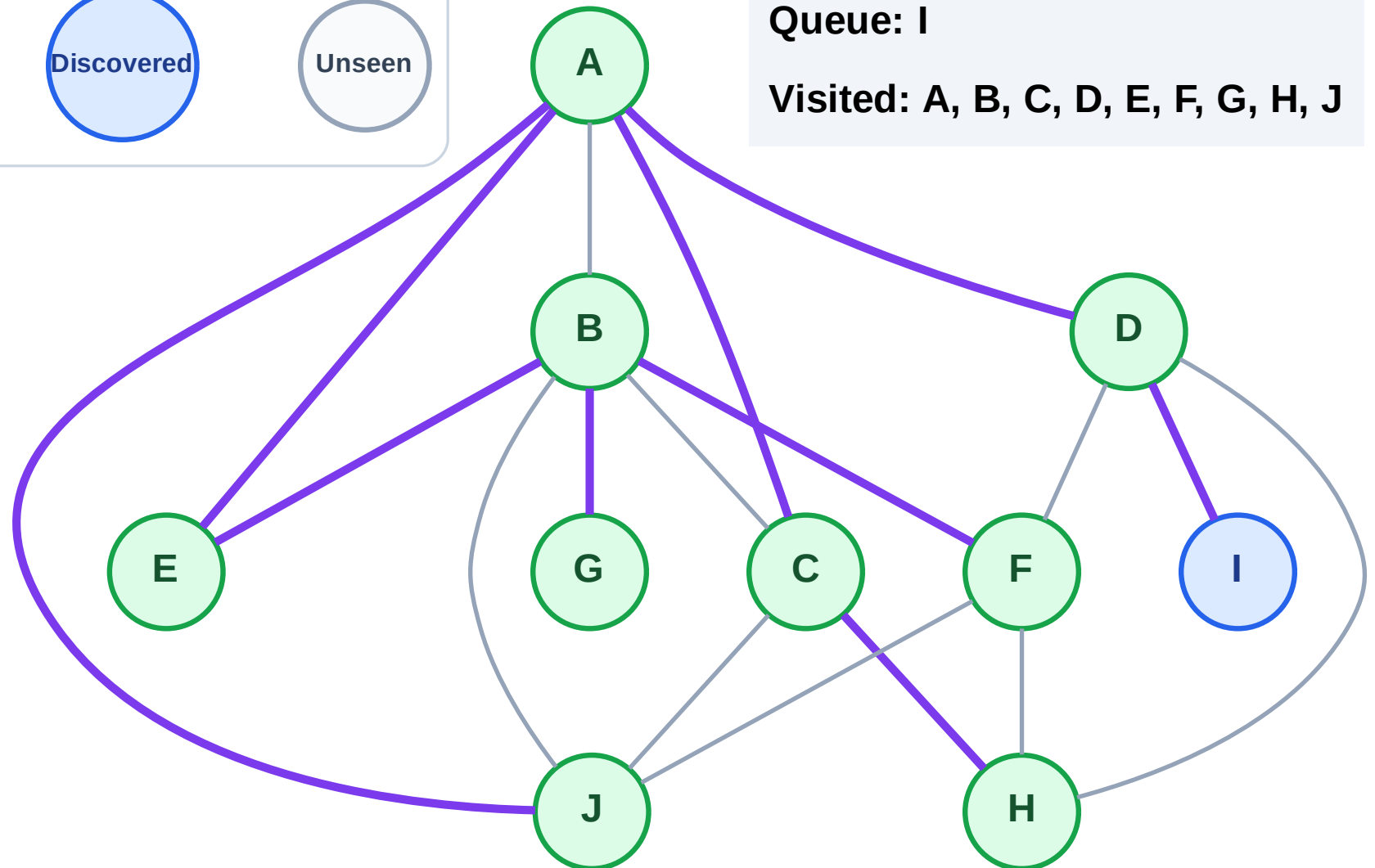
Step 19: No new neighbors from H

Legend



Queue: I

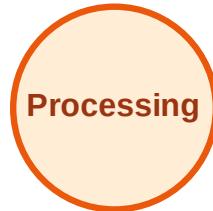
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

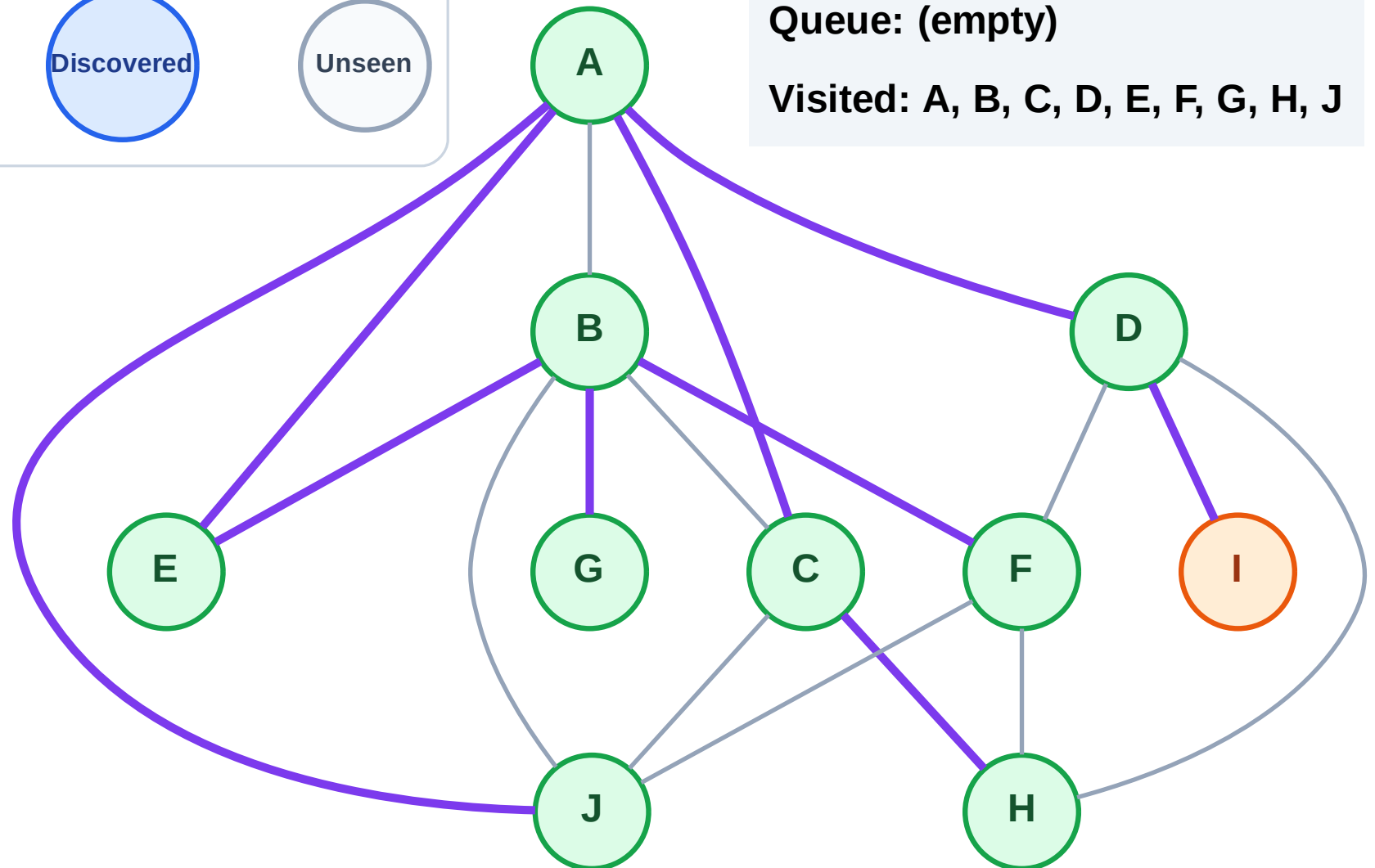
Step 20: Process node I

Legend



Queue: (empty)

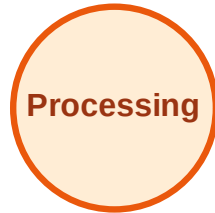
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

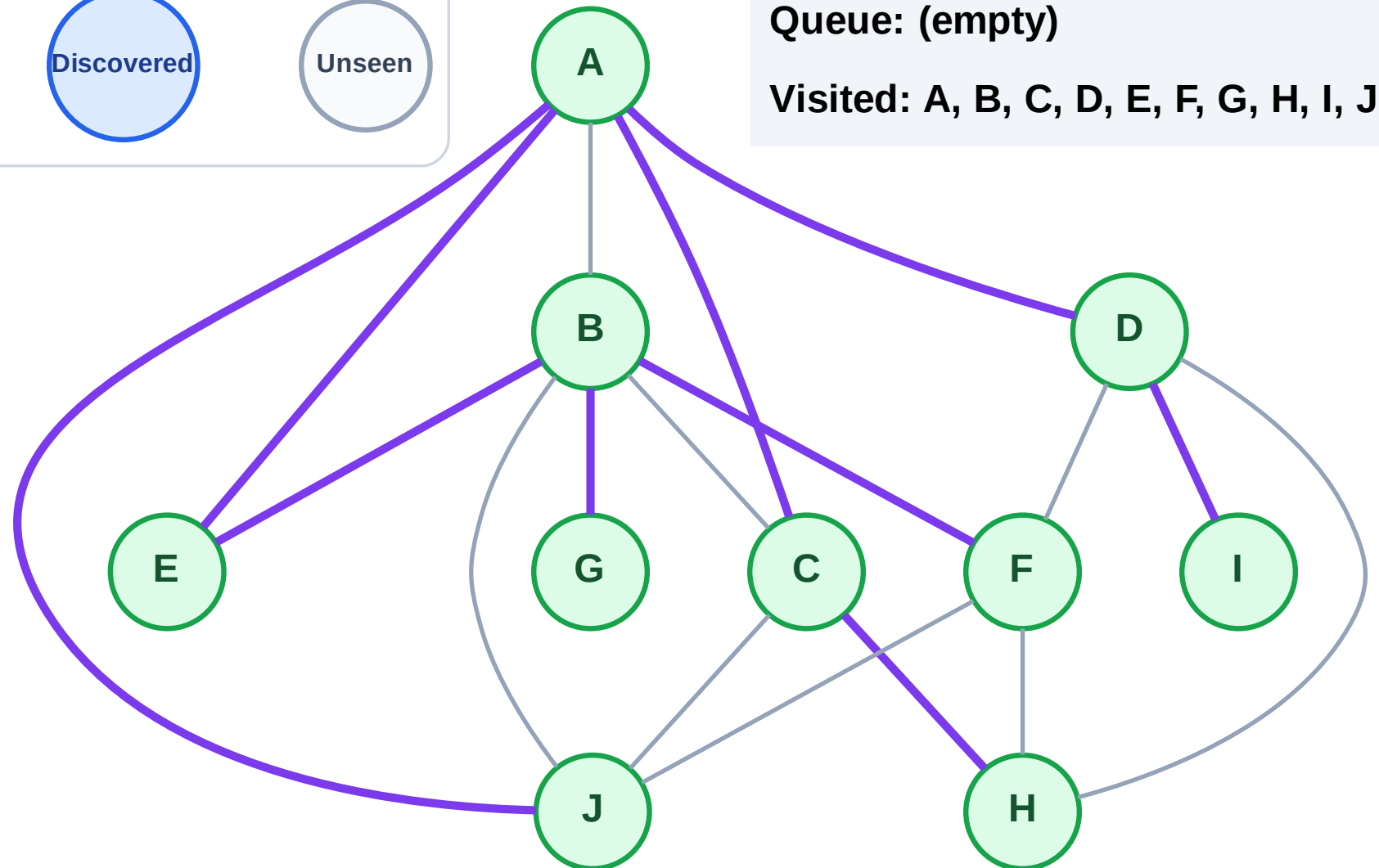
Step 21: No new neighbors from I

Legend



Queue: (empty)

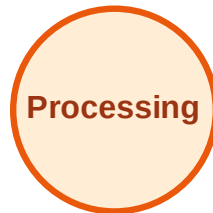
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

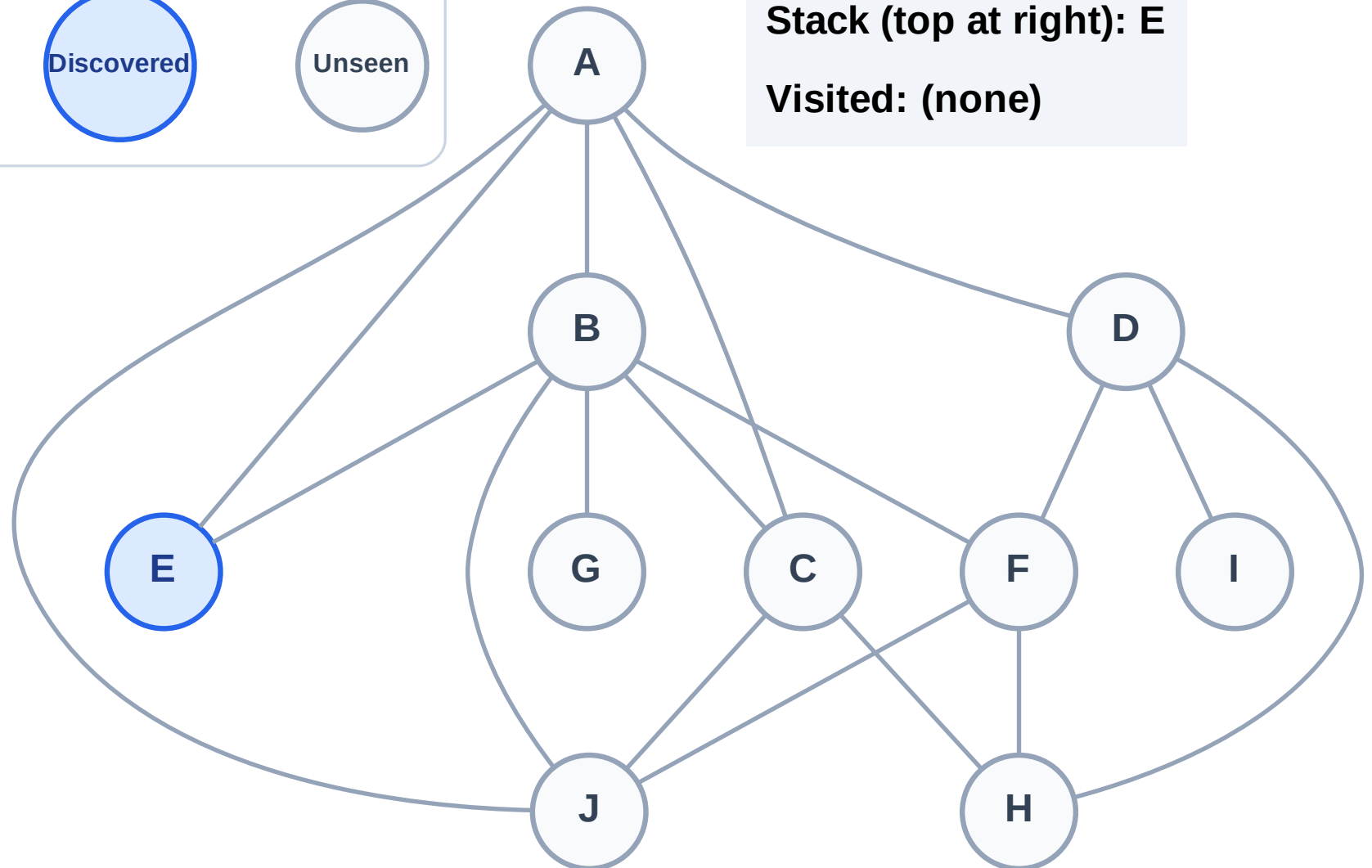
Step 1: Start at E

Legend



Stack (top at right): E

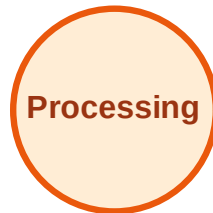
Visited: (none)



DFS Traversal

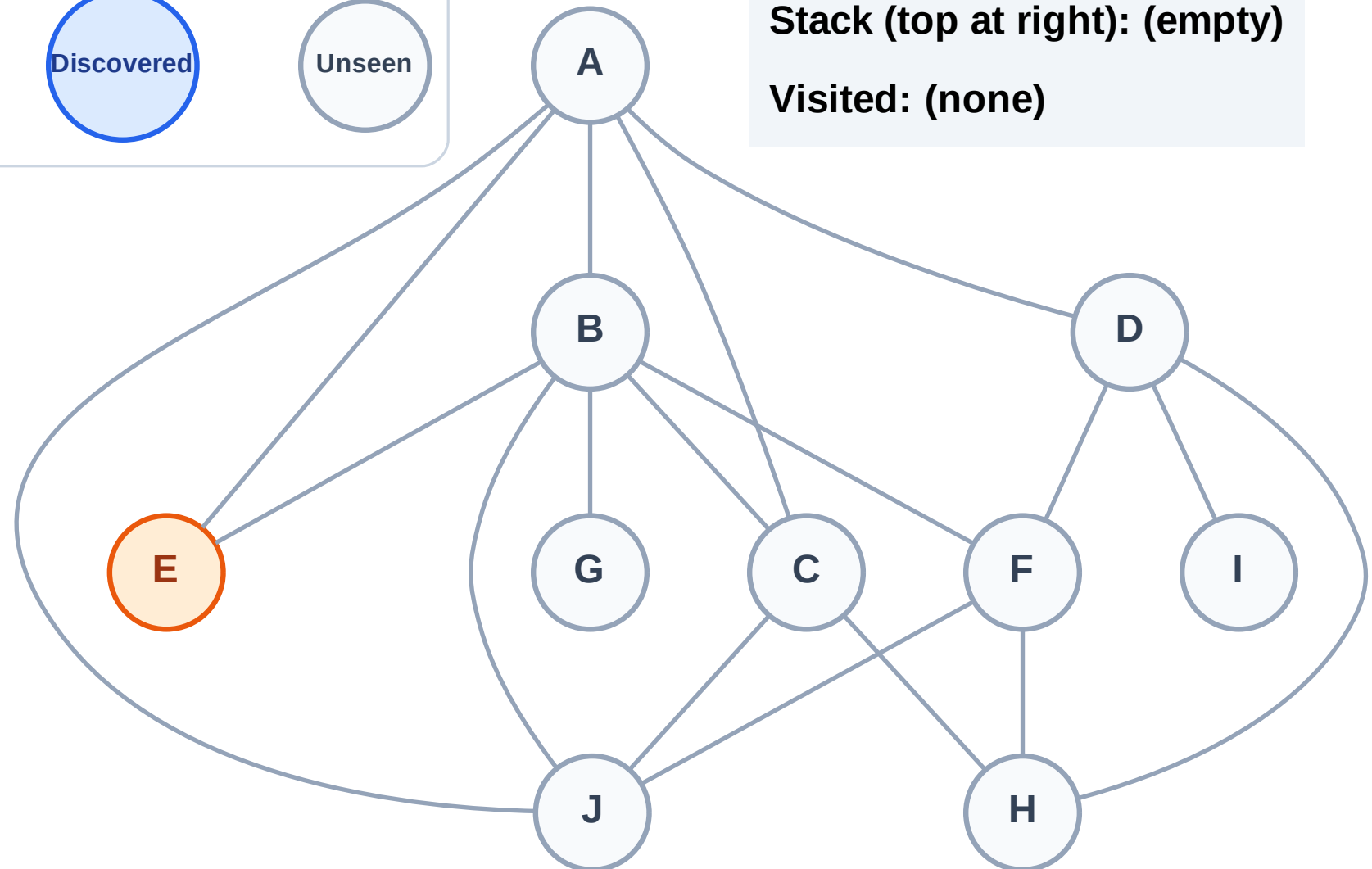
Step 2: Process node E

Legend



Stack (top at right): (empty)

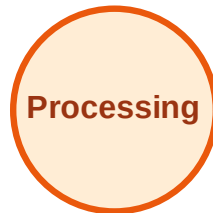
Visited: (none)



DFS Traversal

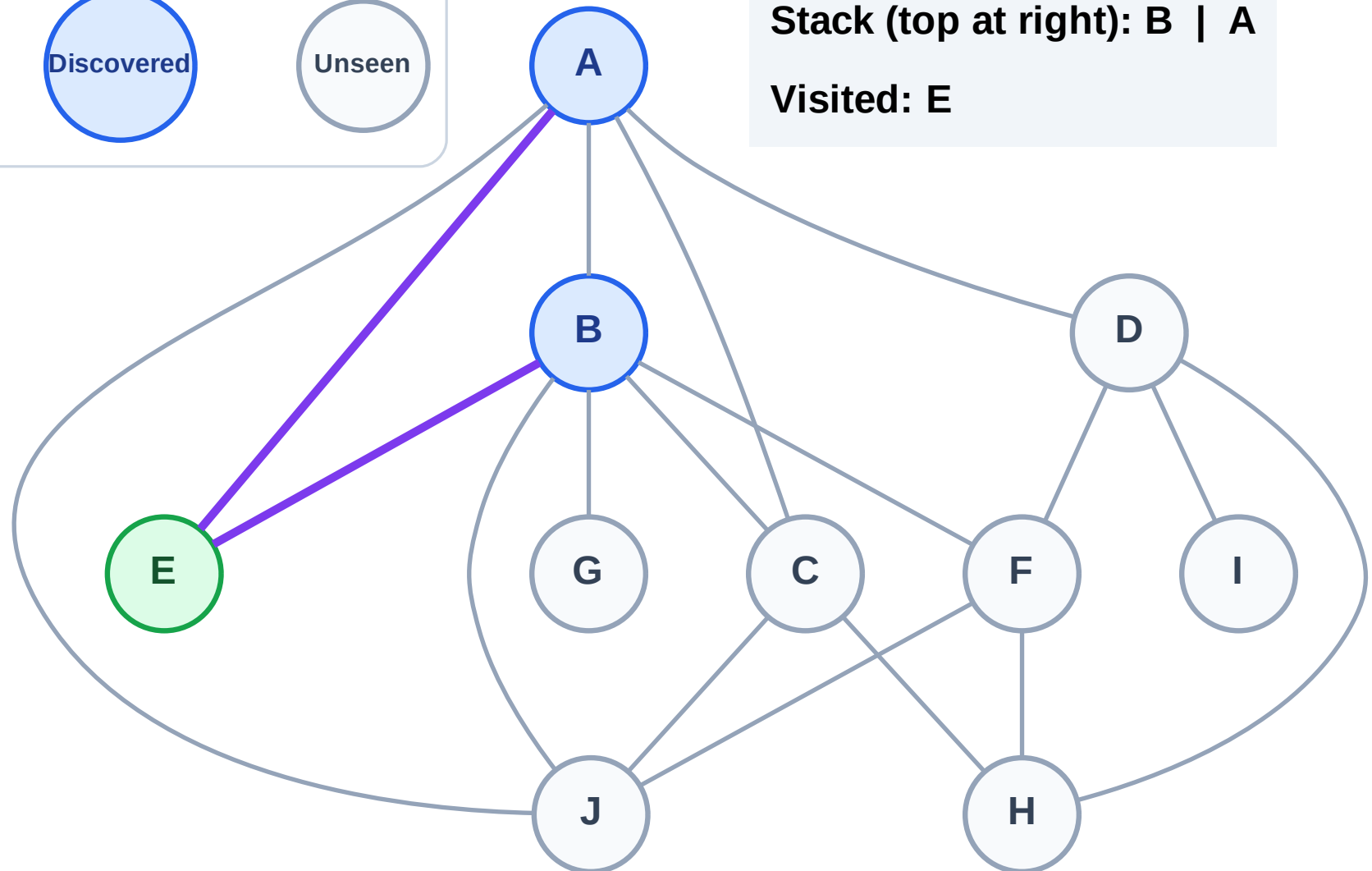
Step 3: Discover B, A from E

Legend



Stack (top at right): B | A

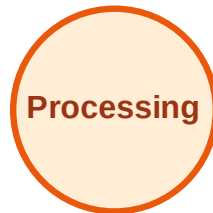
Visited: E



DFS Traversal

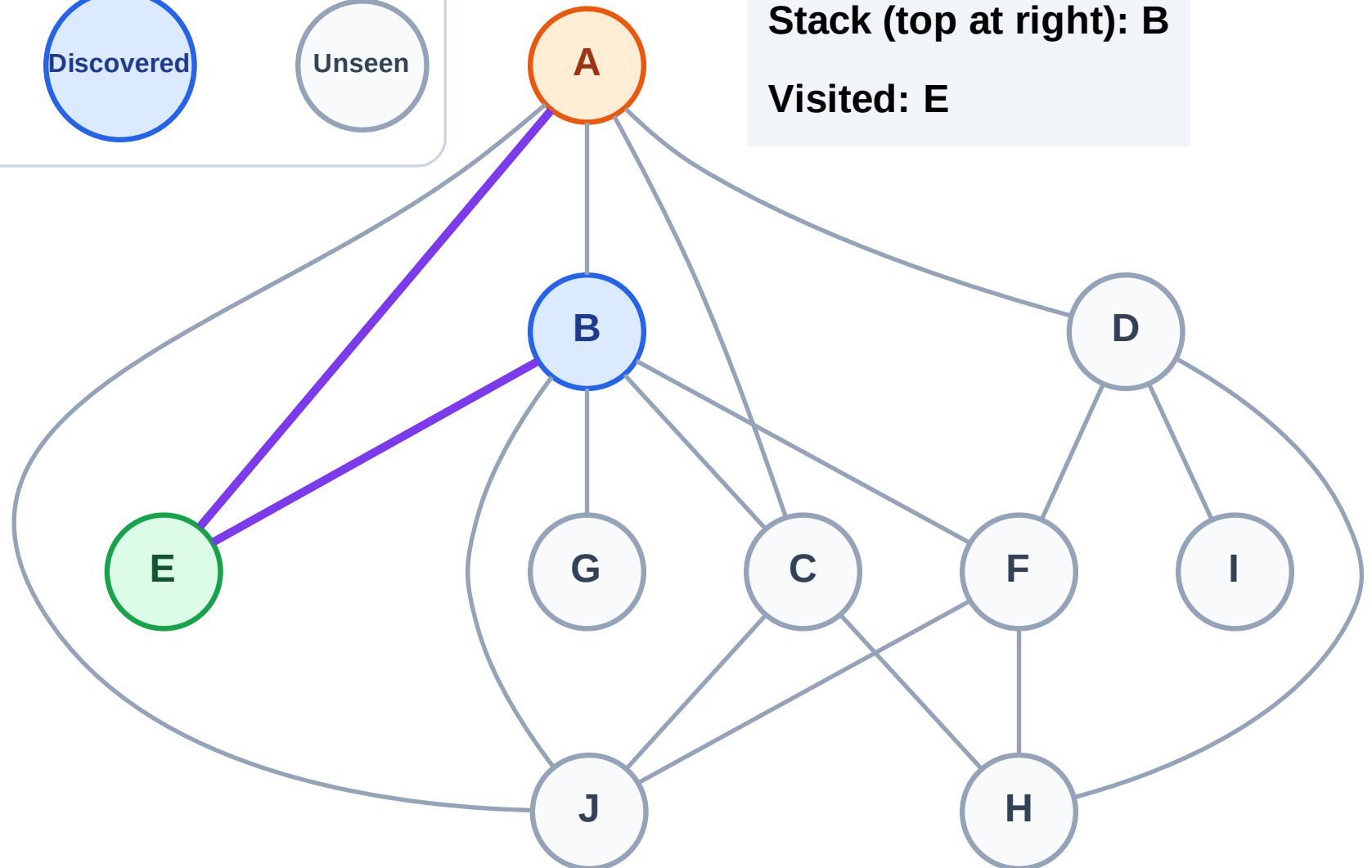
Step 4: Process node A

Legend



Stack (top at right): B

Visited: E



DFS Traversal

Step 5: Discover J, D, C from A

Legend

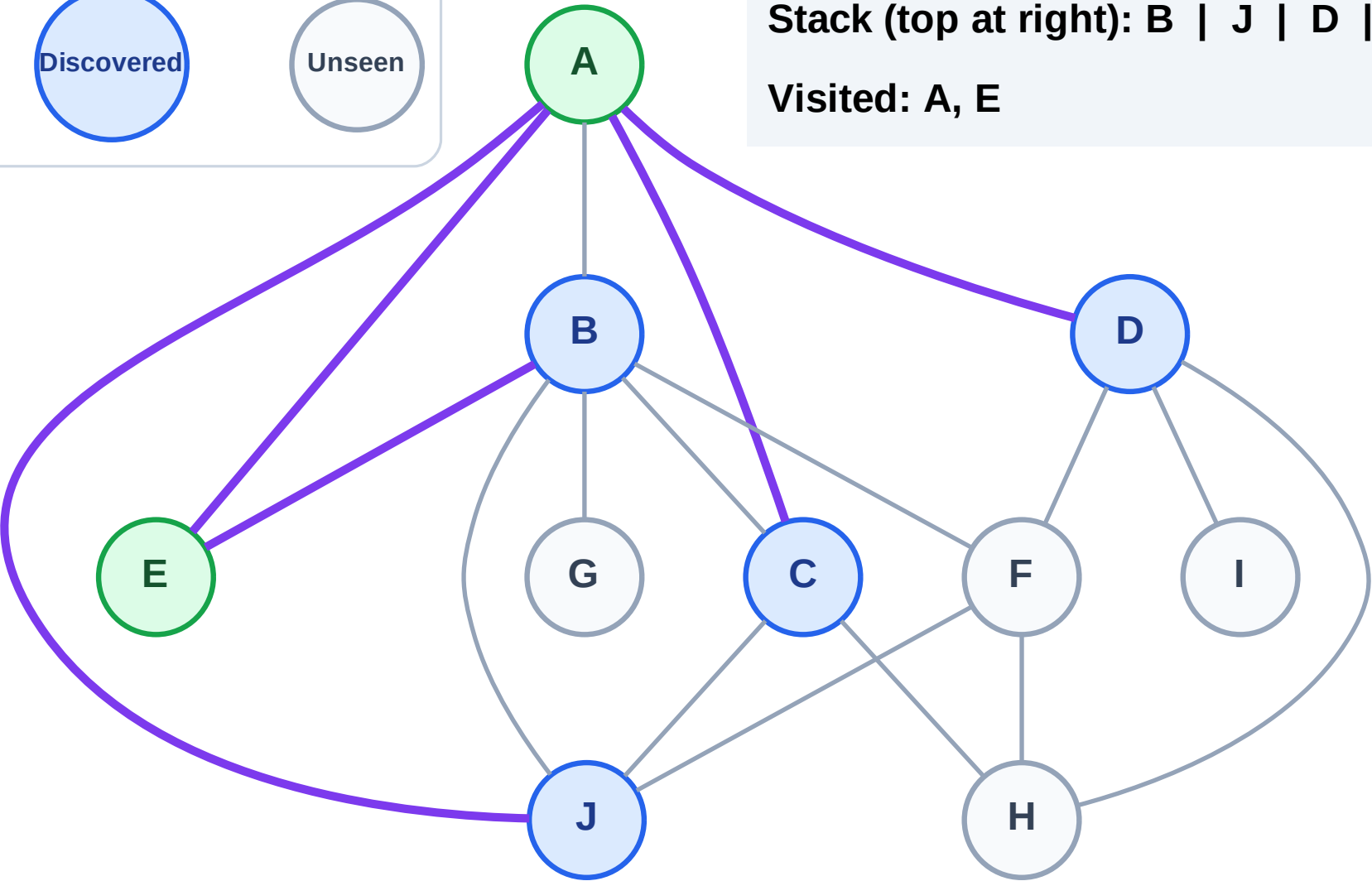
Complete

Processing

Discovered

Unseen

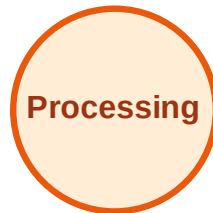
Stack (top at right): B | J | D | C
Visited: A, E



DFS Traversal

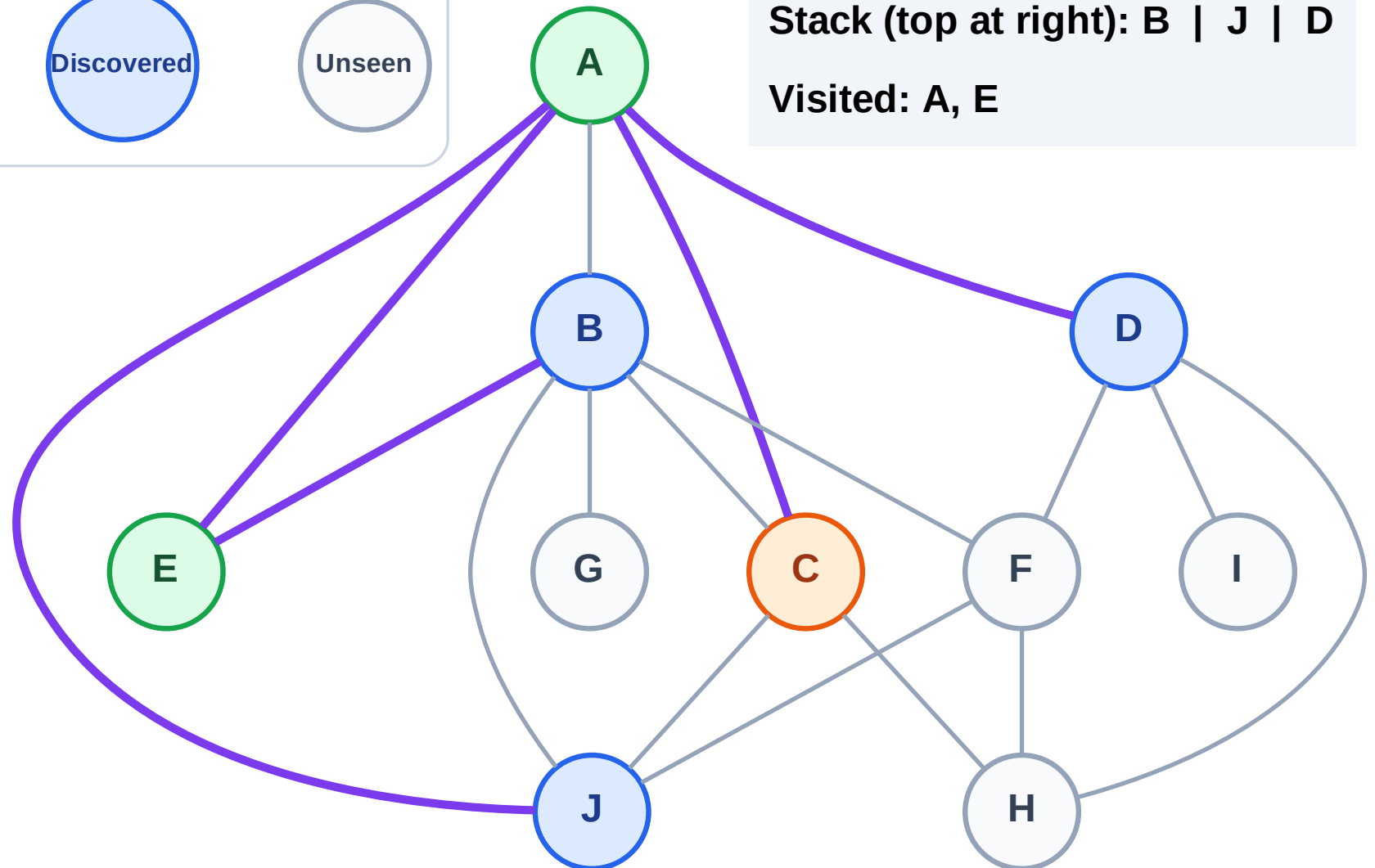
Step 6: Process node C

Legend



Stack (top at right): B | J | D

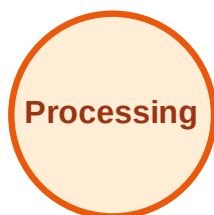
Visited: A, E



DFS Traversal

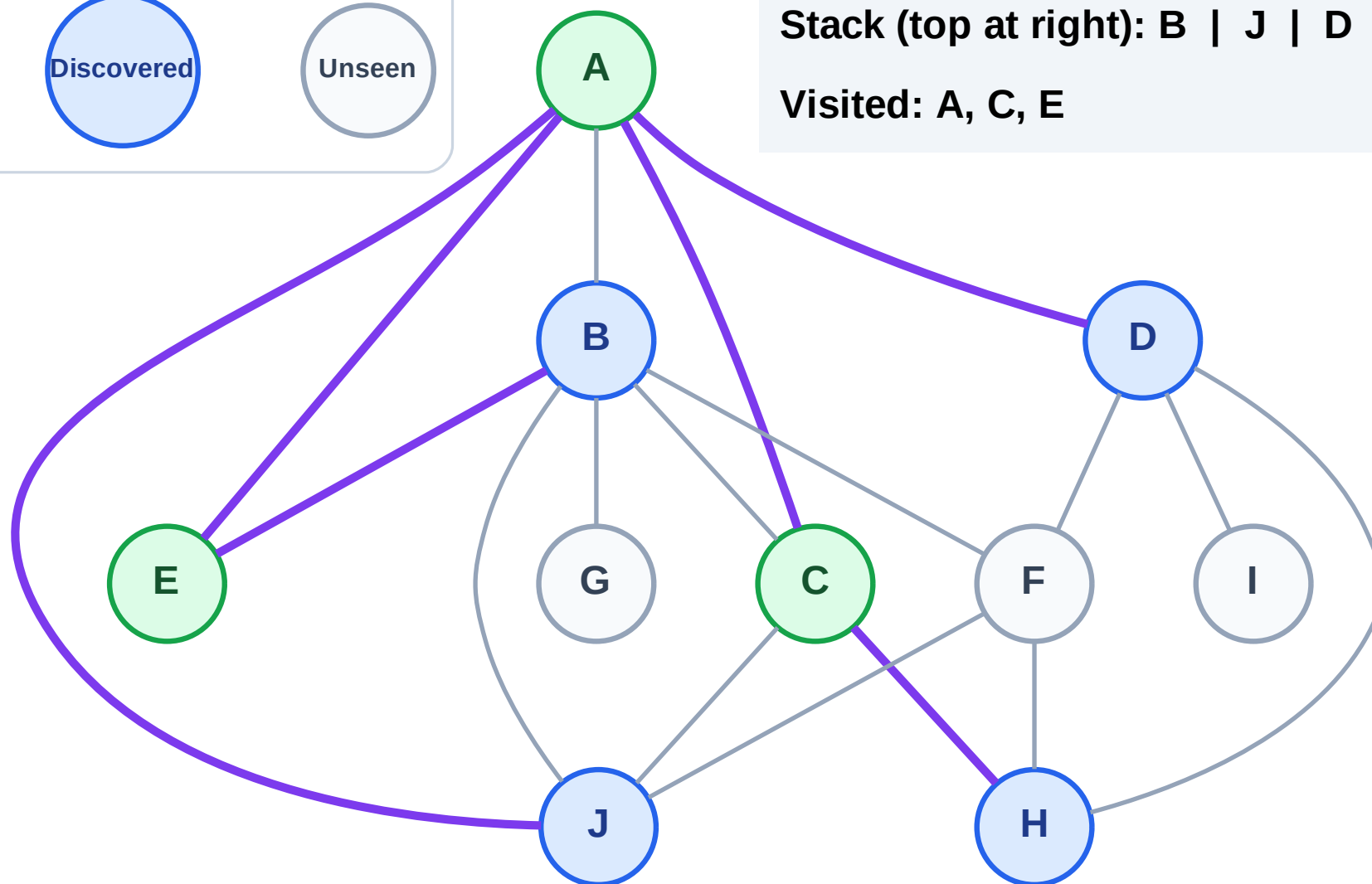
Step 7: Discover H from C

Legend



Stack (top at right): B | J | D | H

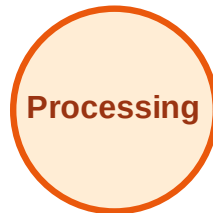
Visited: A, C, E



DFS Traversal

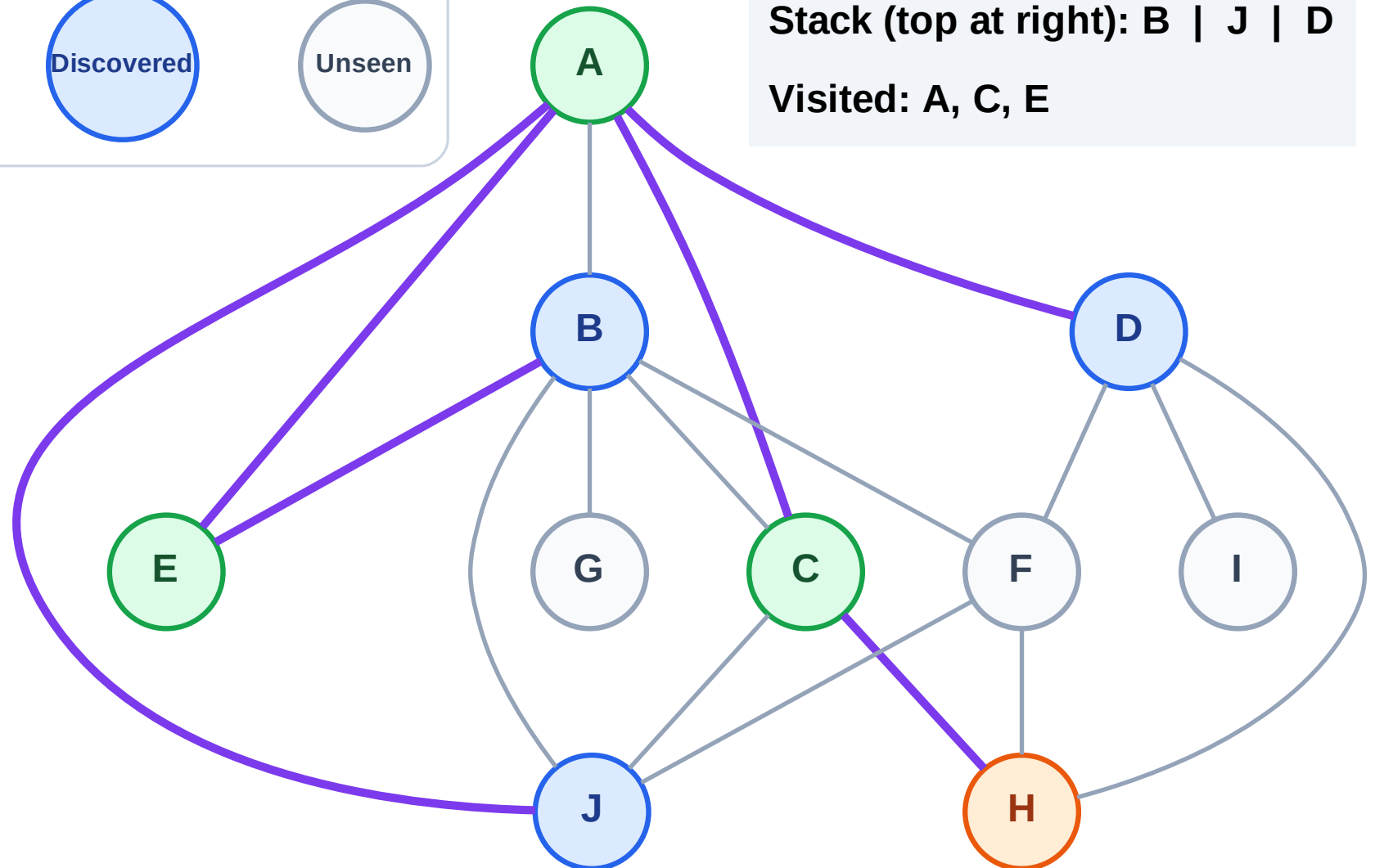
Step 8: Process node H

Legend



Stack (top at right): B | J | D

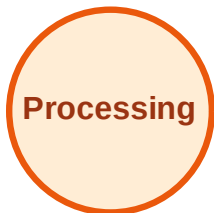
Visited: A, C, E



DFS Traversal

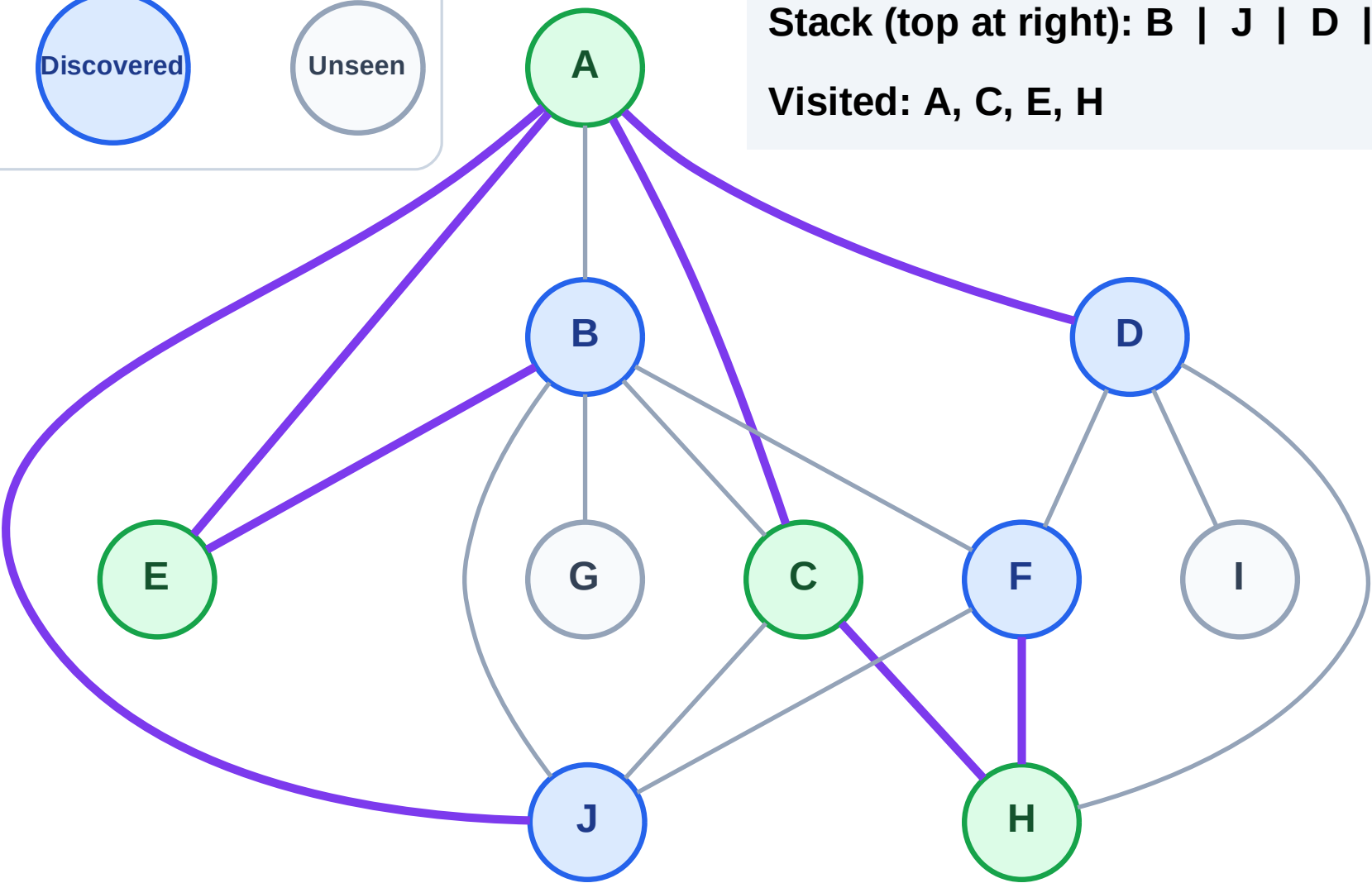
Step 9: Discover F from H

Legend



Stack (top at right): B | J | D | F

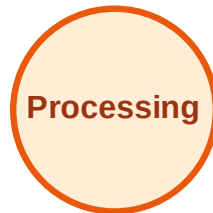
Visited: A, C, E, H



DFS Traversal

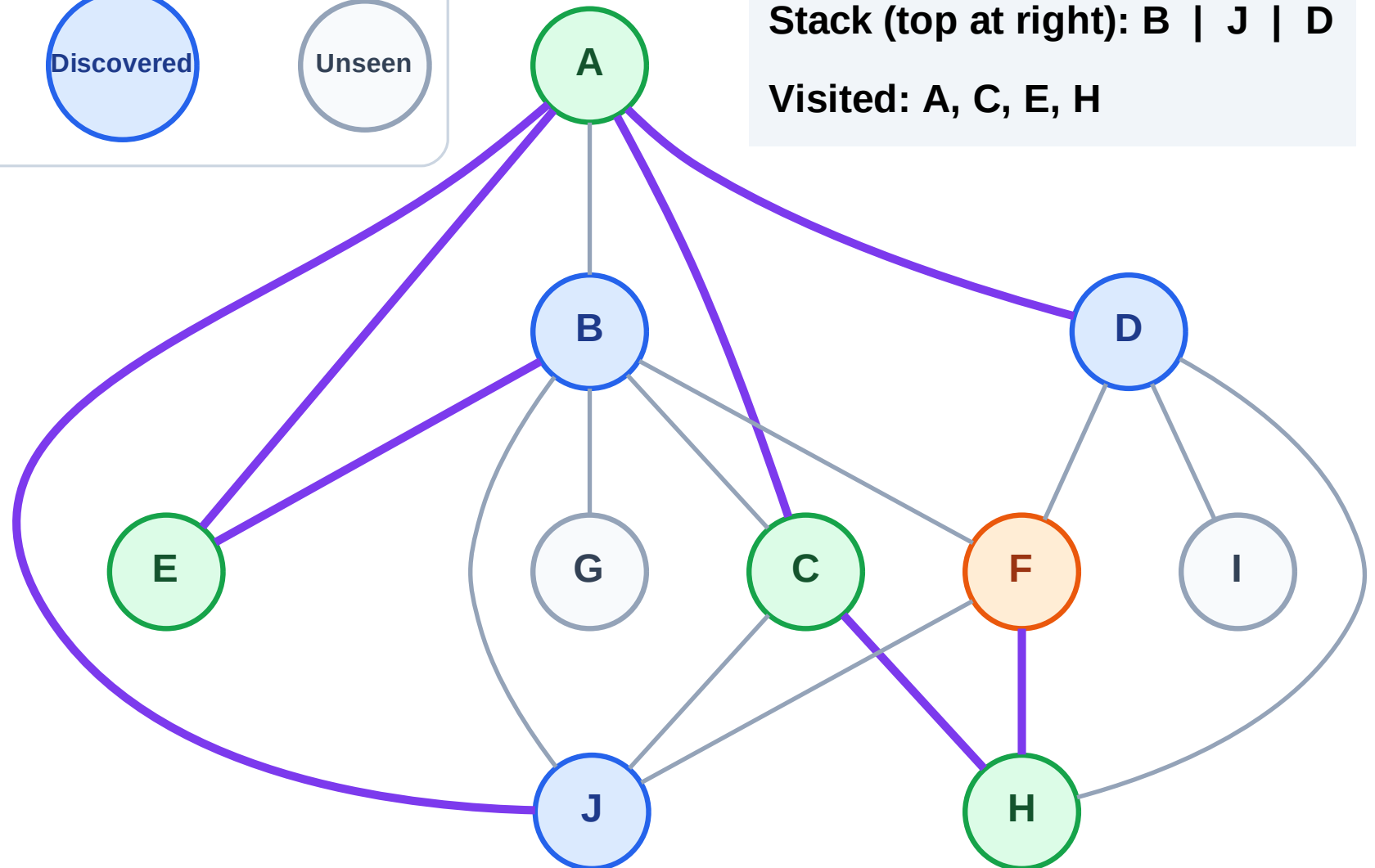
Step 10: Process node F

Legend



Stack (top at right): B | J | D

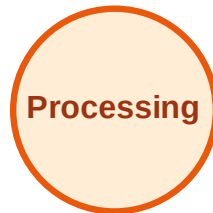
Visited: A, C, E, H



DFS Traversal

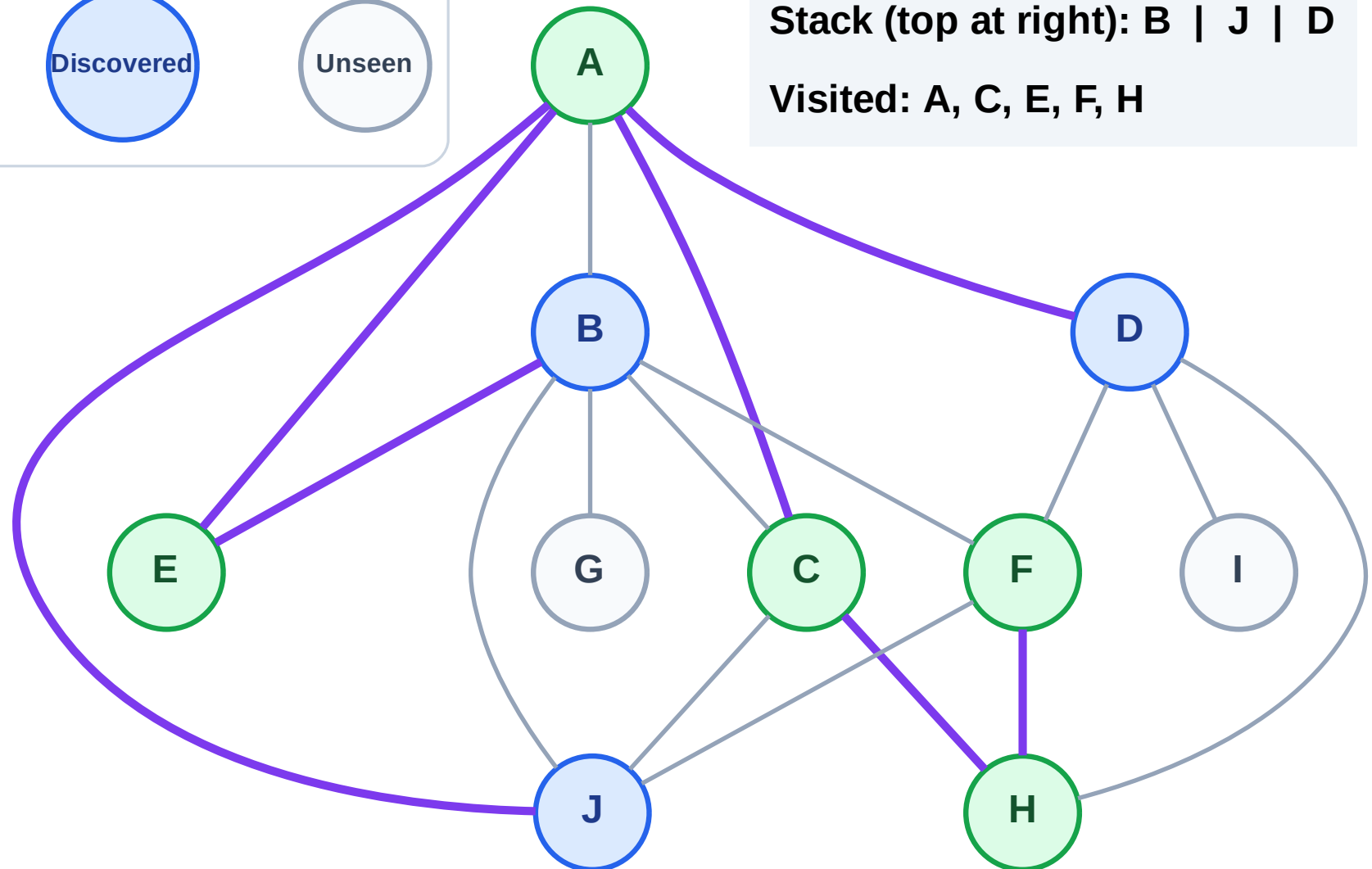
Step 11: No new neighbors from F

Legend



Stack (top at right): B | J | D

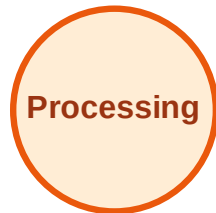
Visited: A, C, E, F, H



DFS Traversal

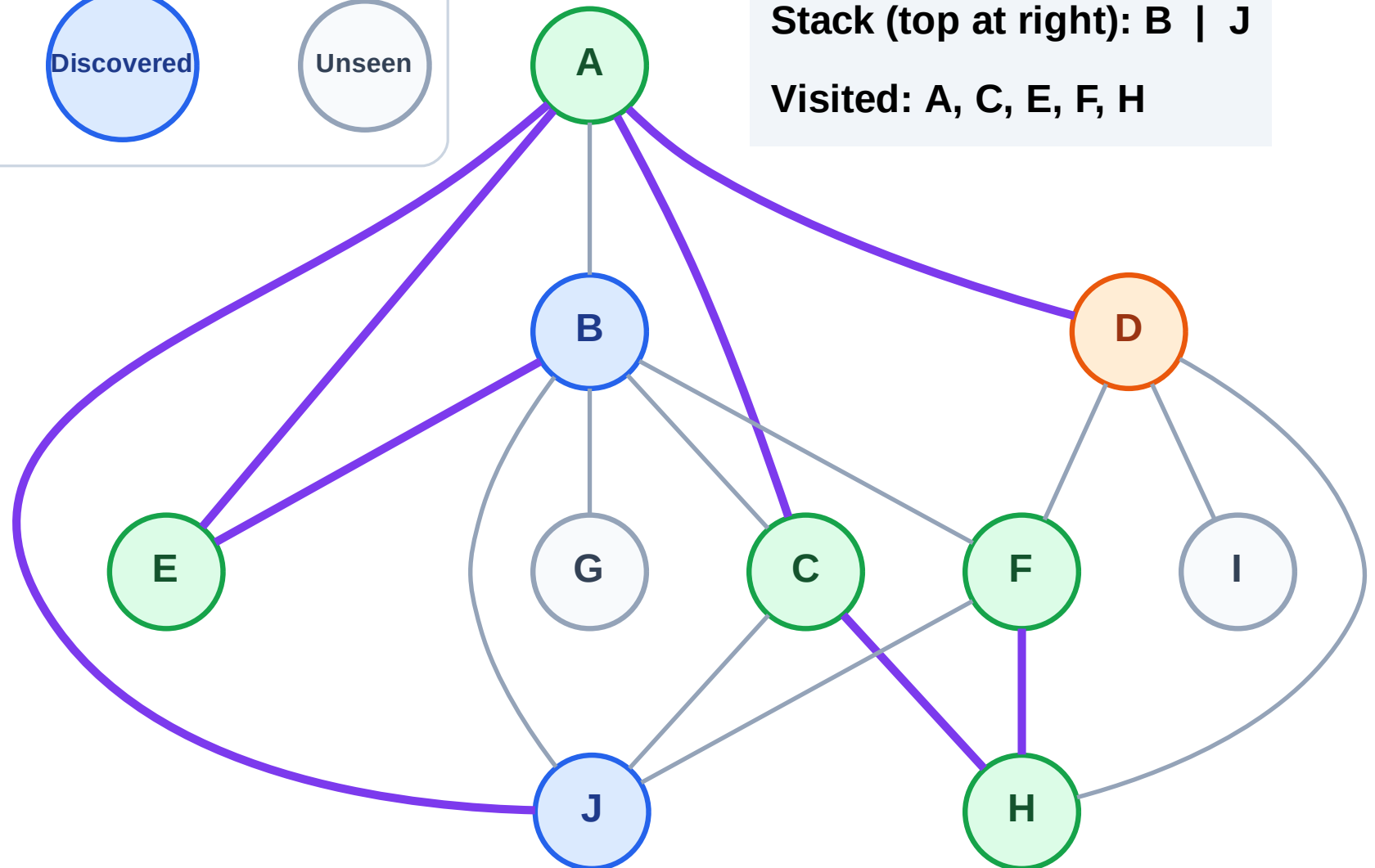
Step 12: Process node D

Legend



Stack (top at right): B | J

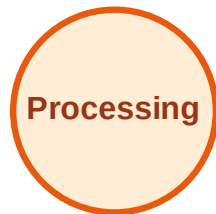
Visited: A, C, E, F, H



DFS Traversal

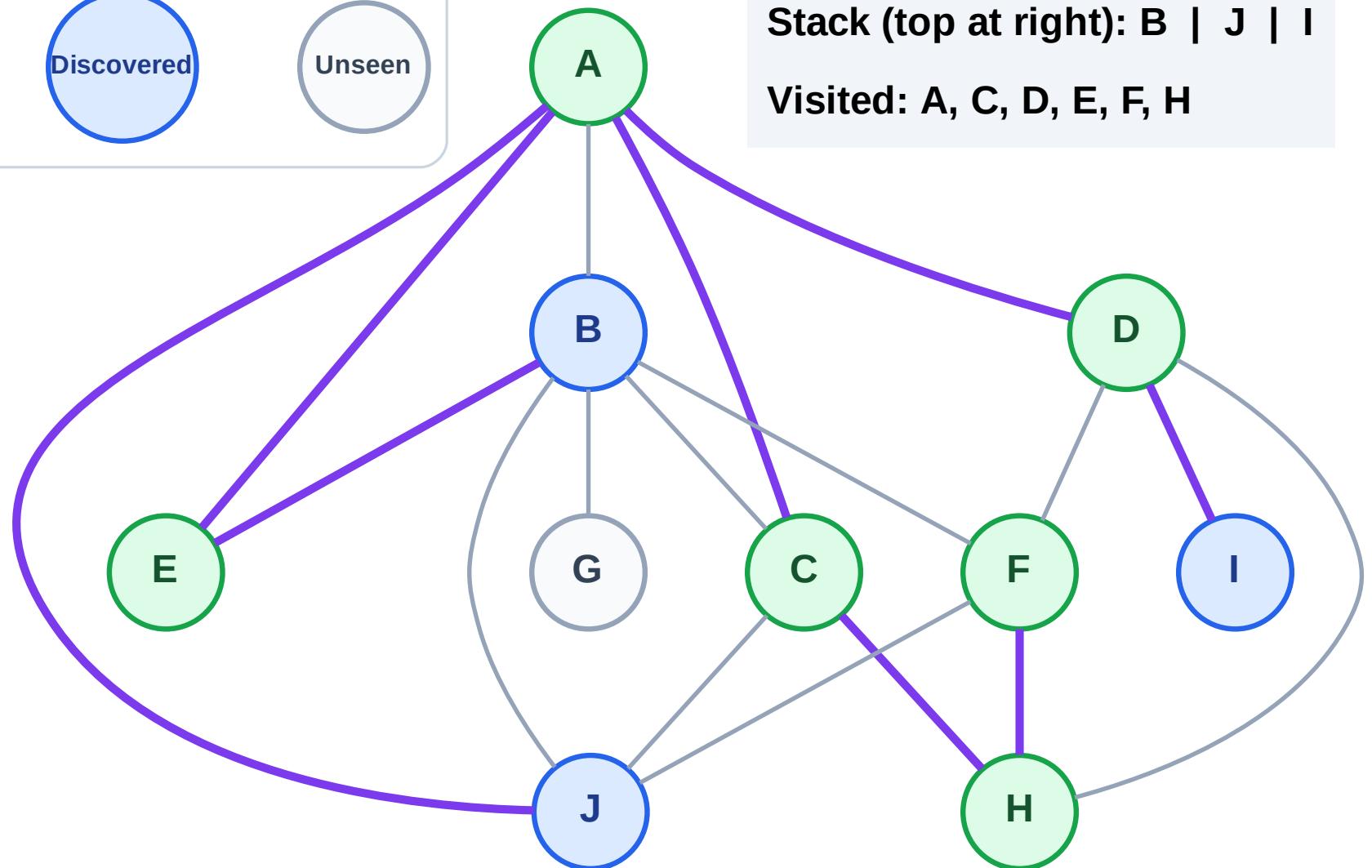
Step 13: Discover I from D

Legend



Stack (top at right): B | J | I

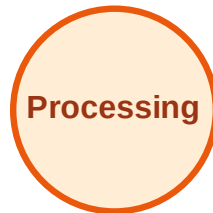
Visited: A, C, D, E, F, H



DFS Traversal

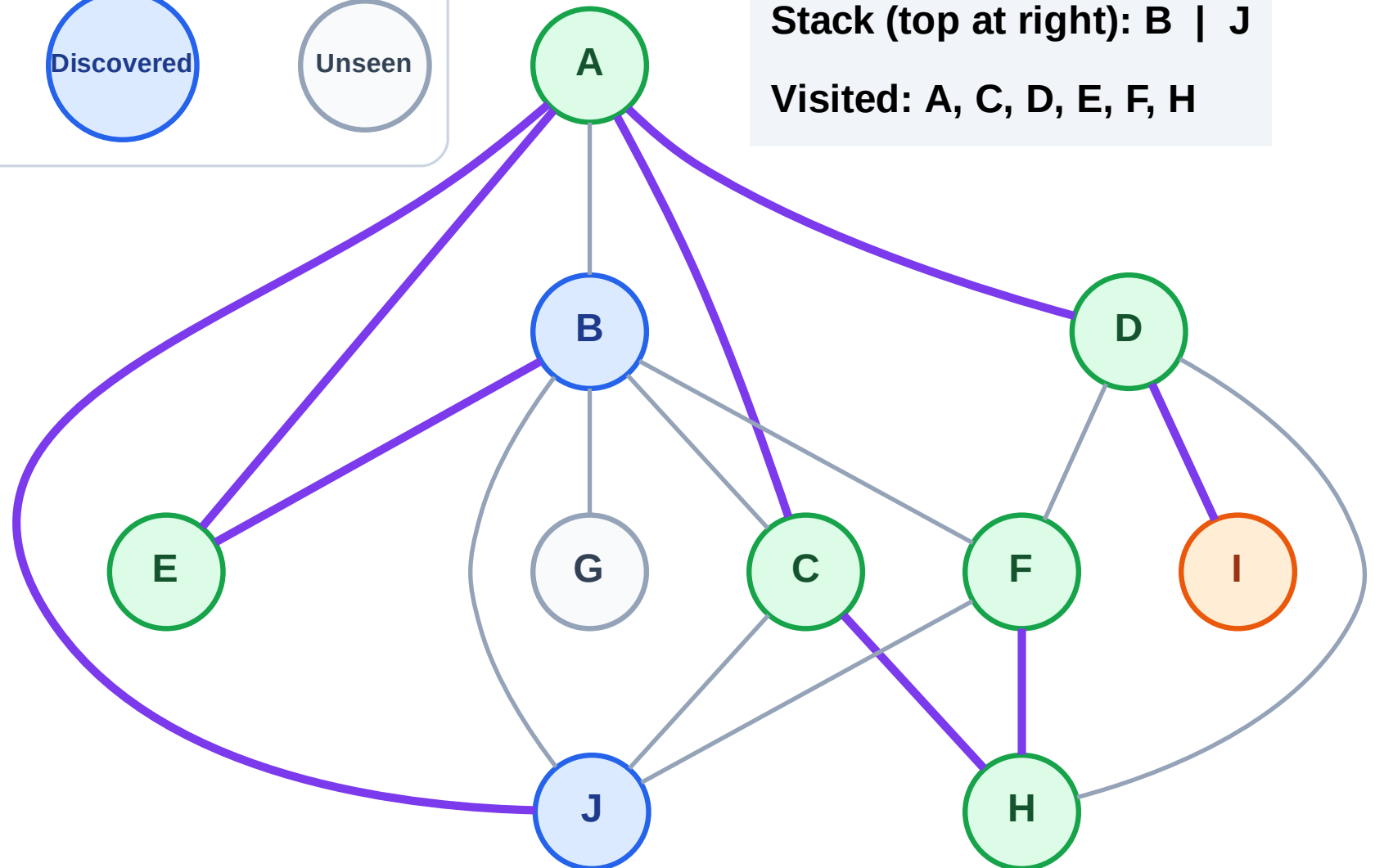
Step 14: Process node I

Legend



Stack (top at right): B | J

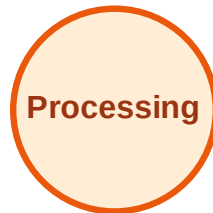
Visited: A, C, D, E, F, H



DFS Traversal

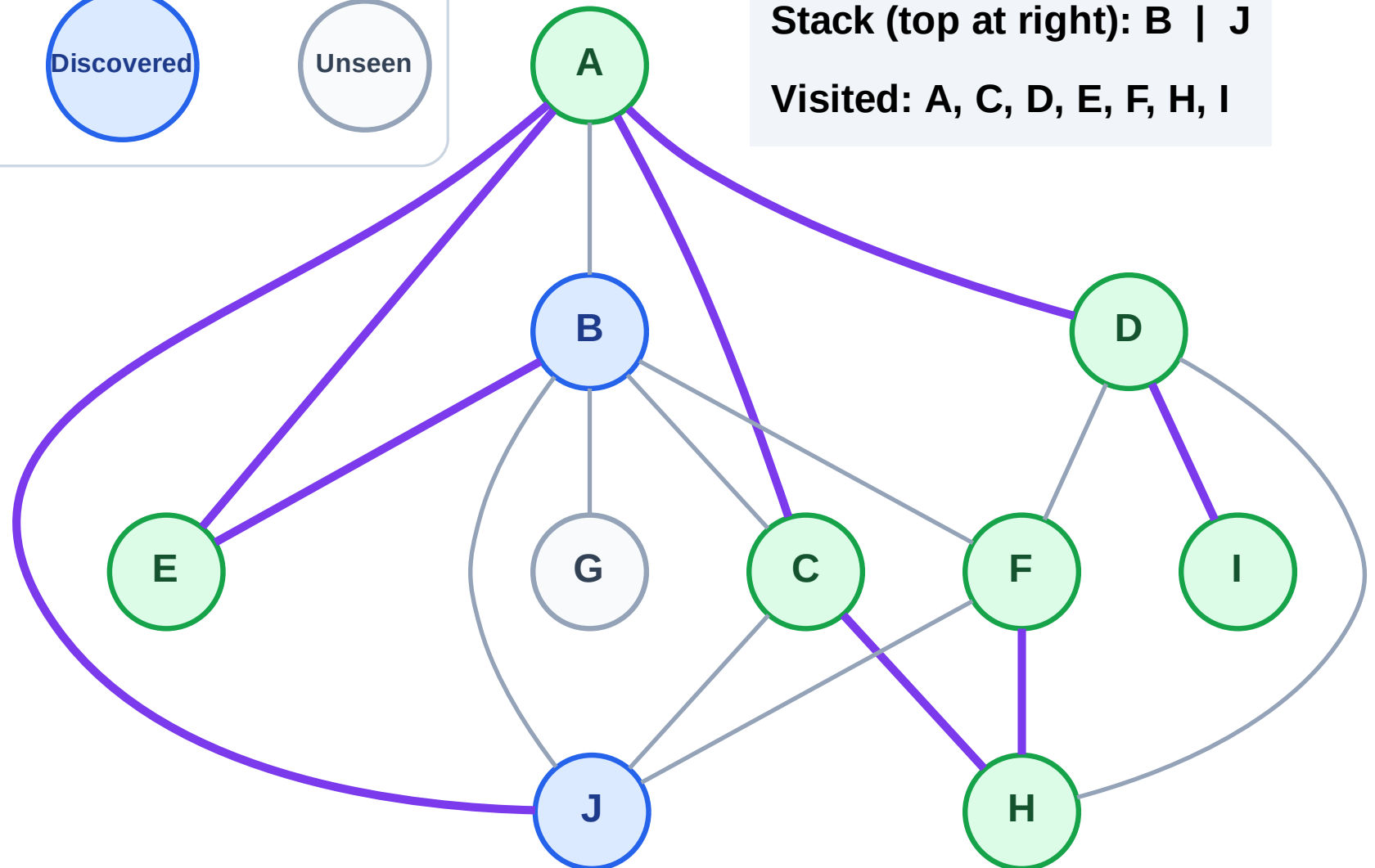
Step 15: No new neighbors from I

Legend



Stack (top at right): B | J

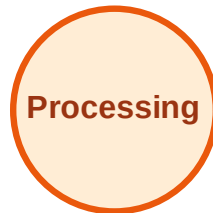
Visited: A, C, D, E, F, H, I



DFS Traversal

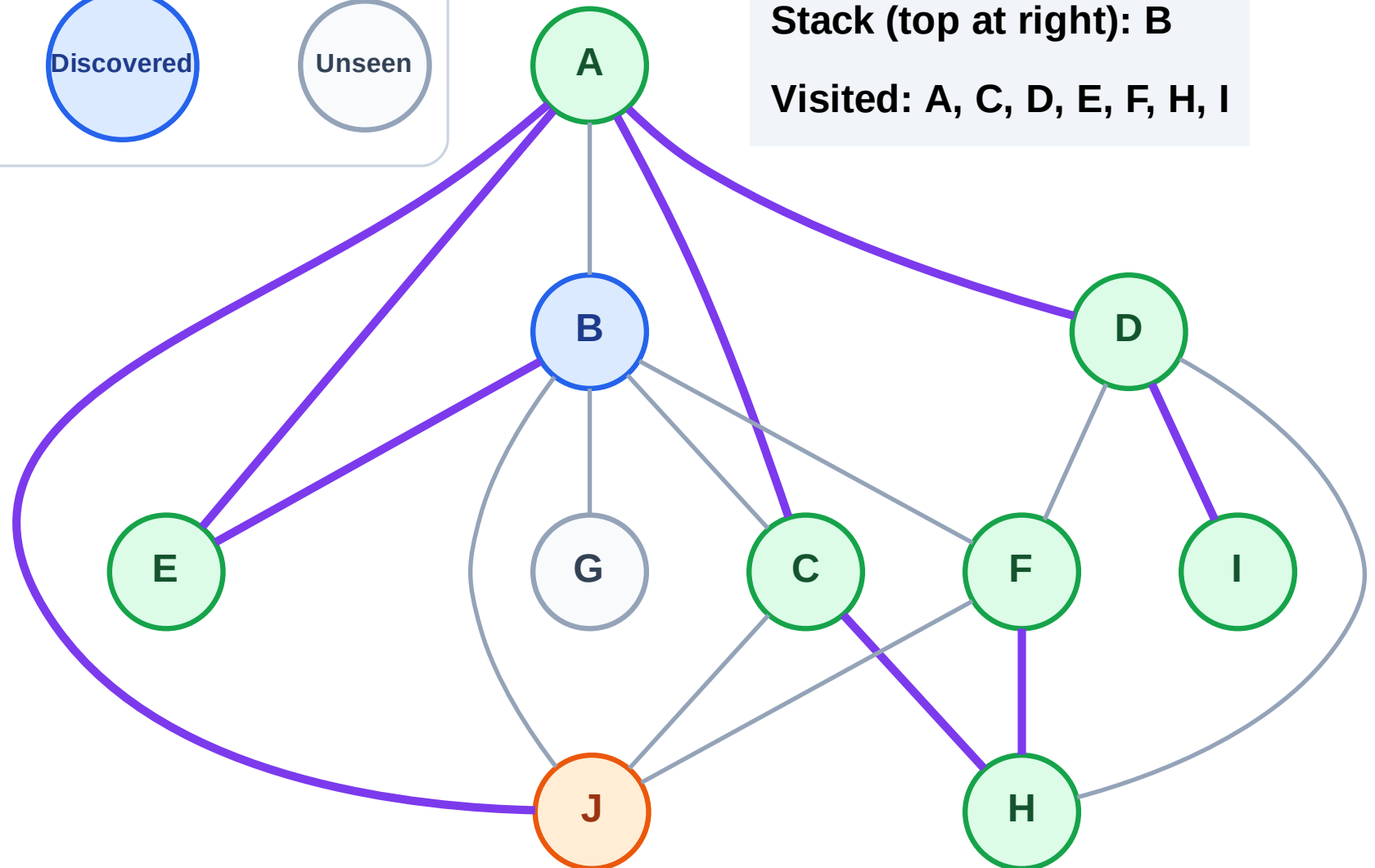
Step 16: Process node J

Legend



Stack (top at right): B

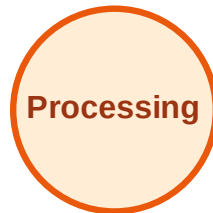
Visited: A, C, D, E, F, H, I



DFS Traversal

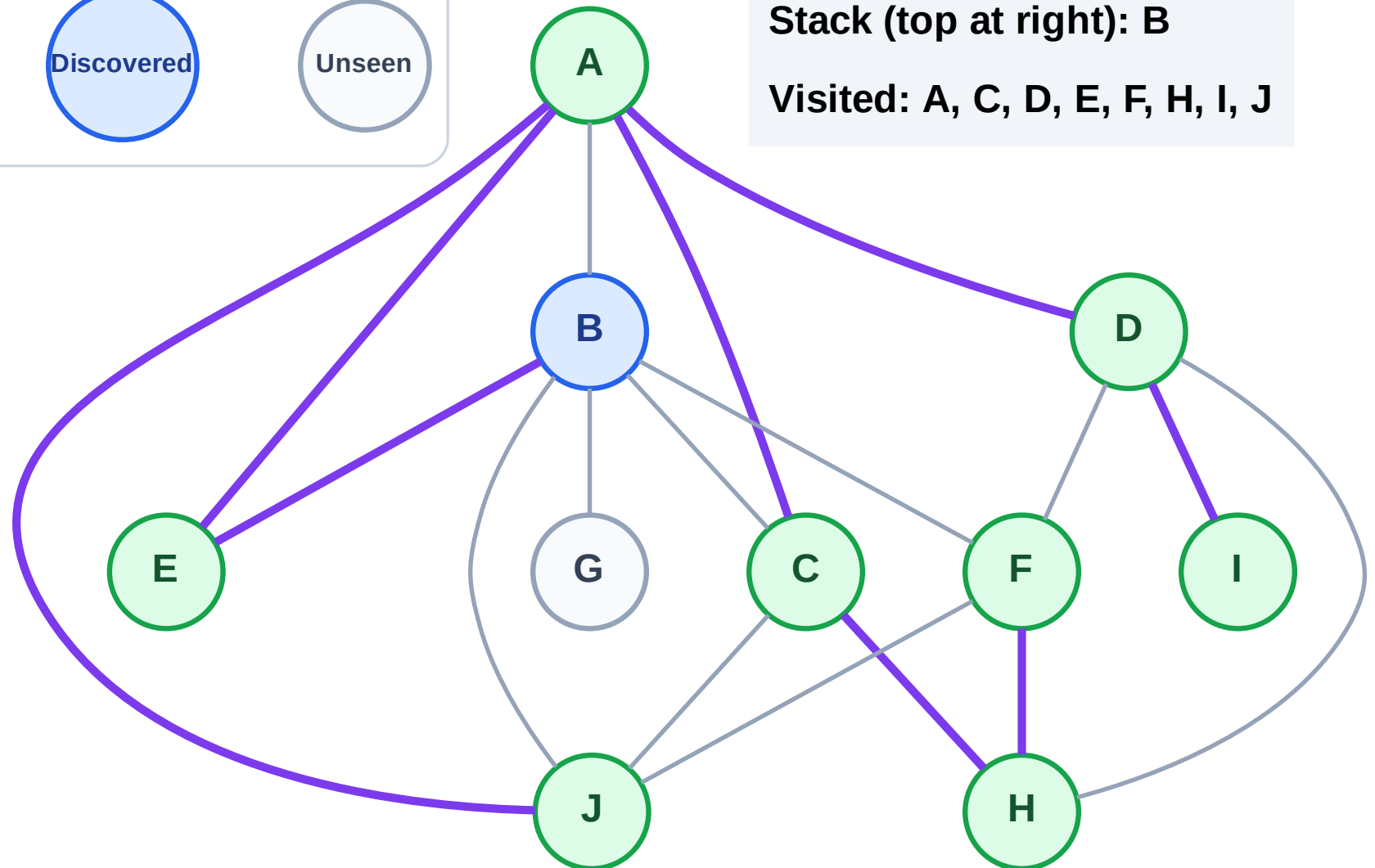
Step 17: No new neighbors from J

Legend



Stack (top at right): B

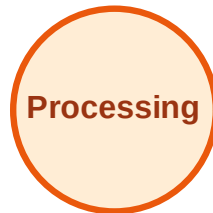
Visited: A, C, D, E, F, H, I, J



DFS Traversal

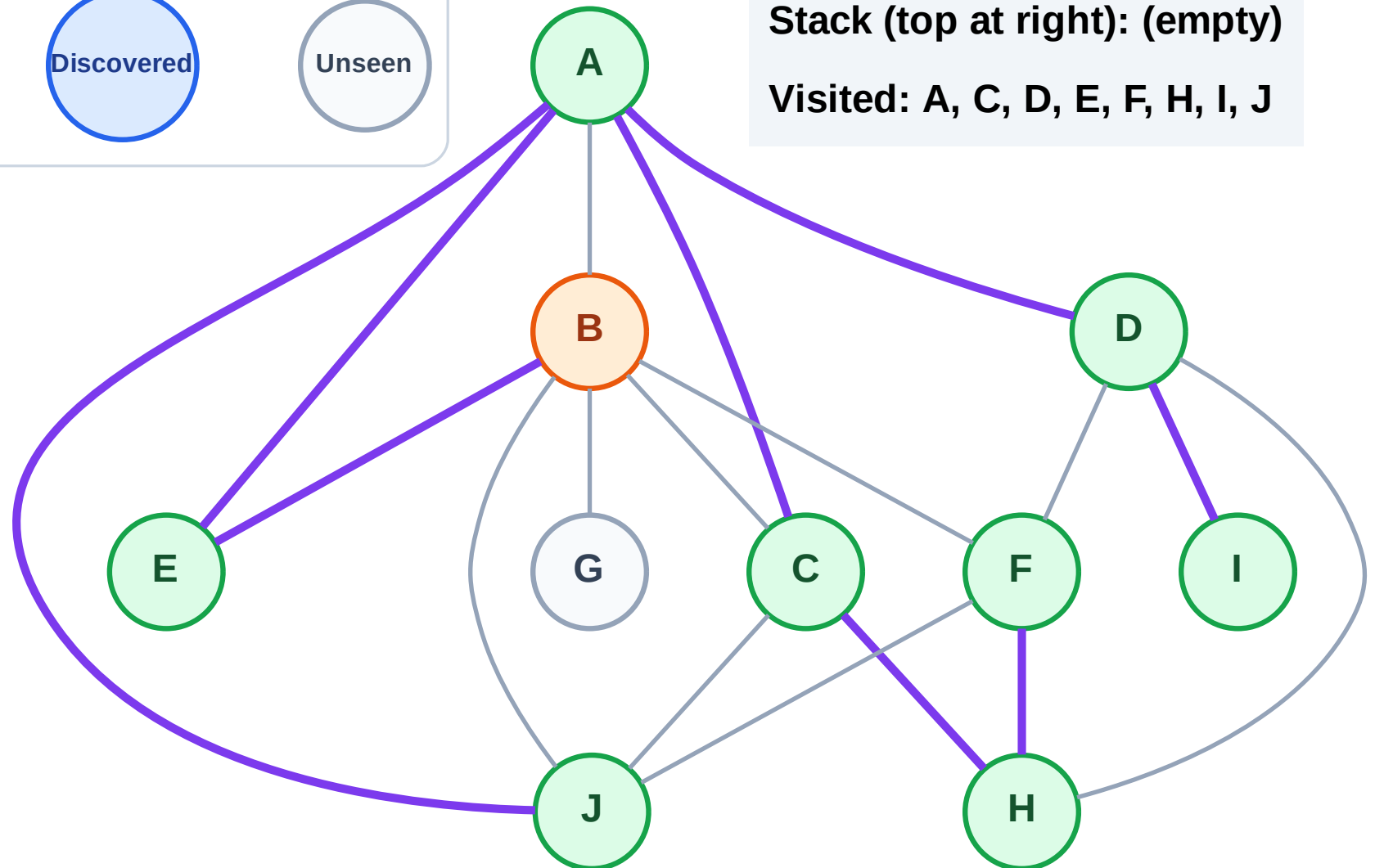
Step 18: Process node B

Legend



Stack (top at right): (empty)

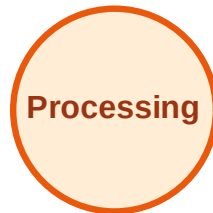
Visited: A, C, D, E, F, H, I, J



DFS Traversal

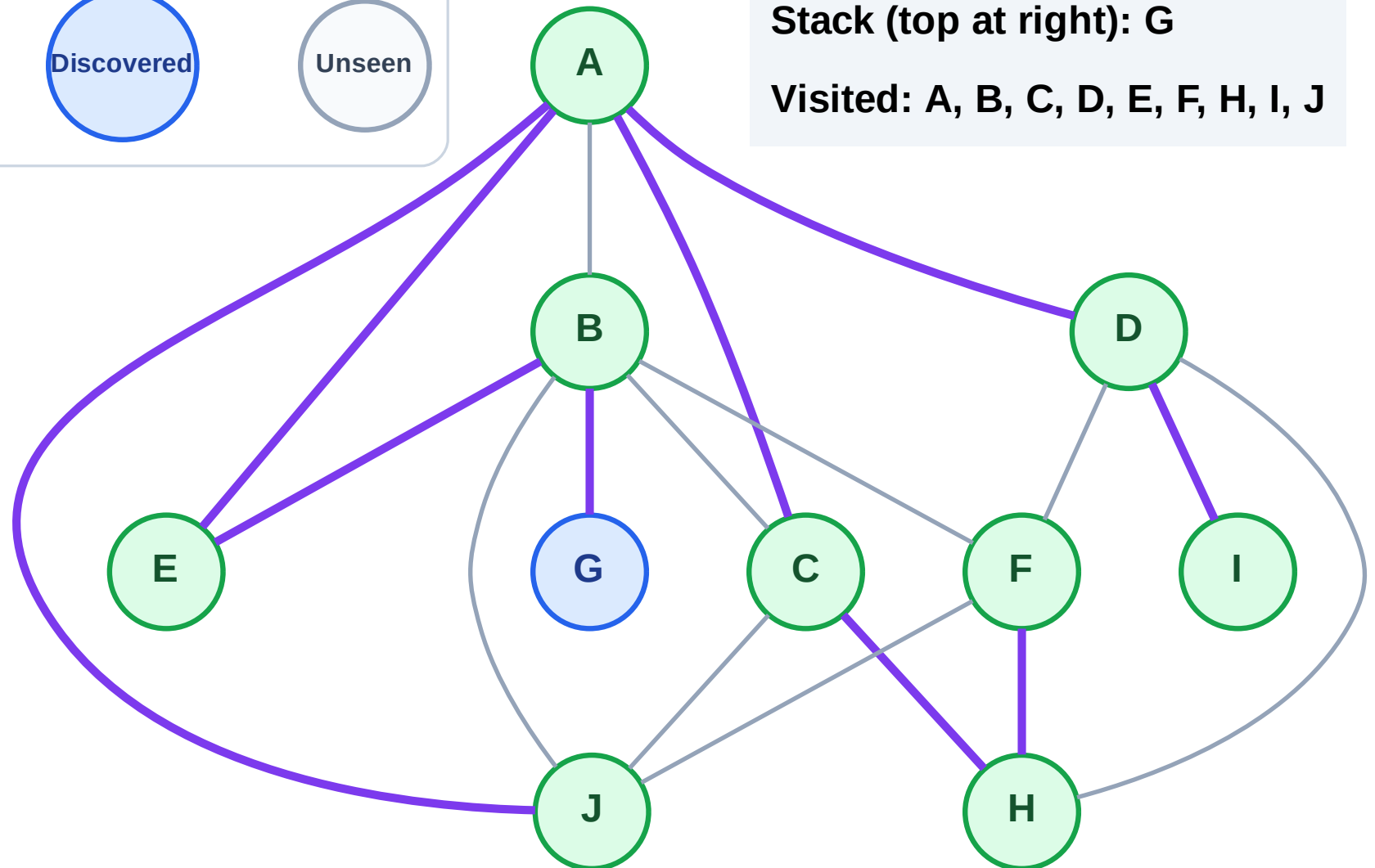
Step 19: Discover G from B

Legend



Stack (top at right): G

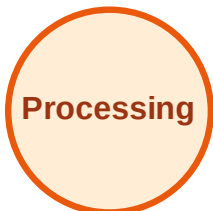
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

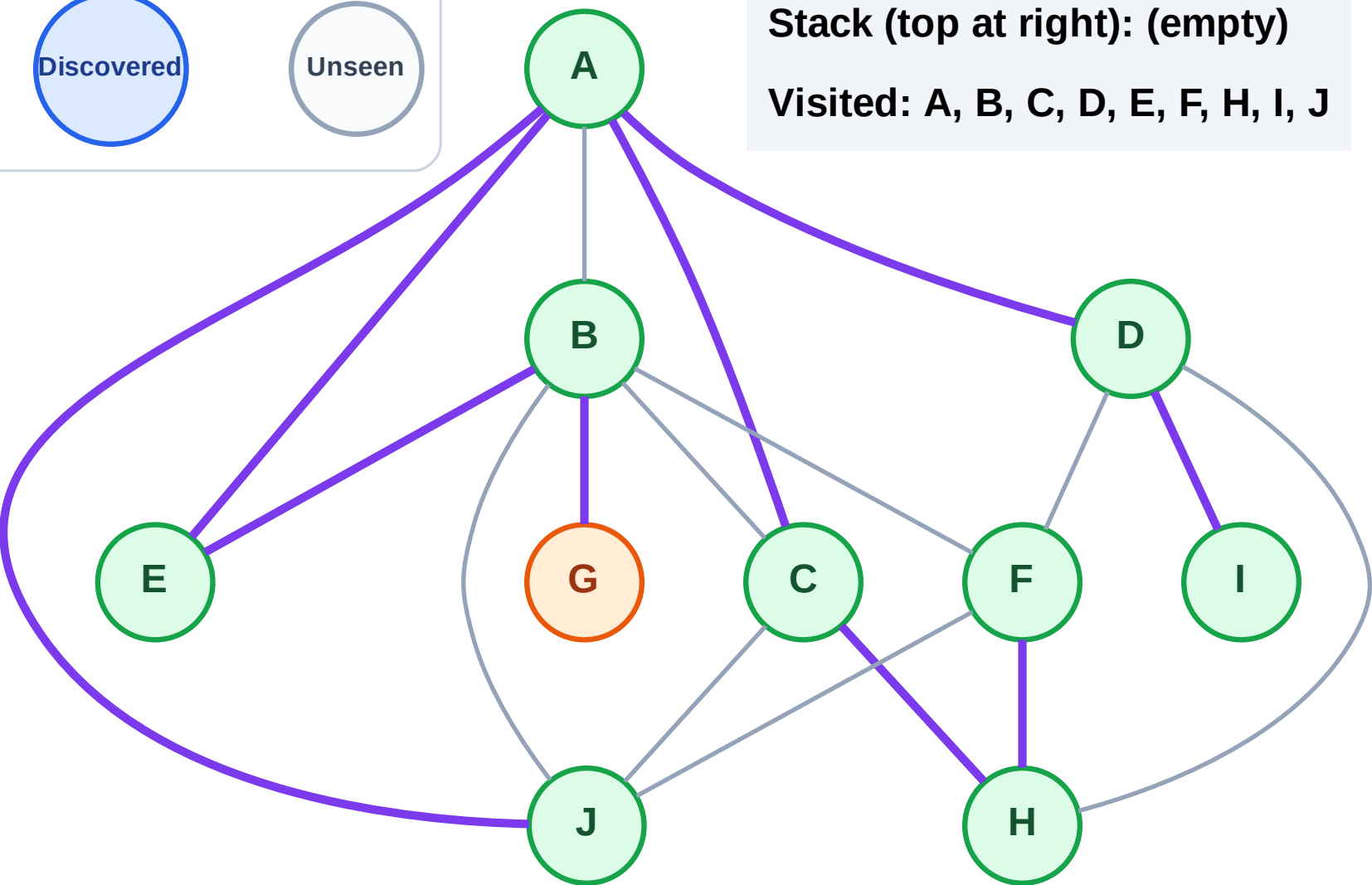
Step 20: Process node G

Legend



Stack (top at right): (empty)

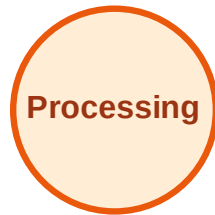
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

